



M. B. M. UNIVERSITY JODHPUR

Date 4/5/23Experiment No. 8P. No. 59Name Zahid KhanFaculty No. 20/24765Class BE III SemBatch Asy

* The two devices are used to reduce pollutions are:-

EGR (Exhaust gas Recirculation):-

- Used for Diesel & Petrol Engines.

- Excessive NO_x forms when peak temp^s exceeds 1927°C .

To lower the combustion temp^s many engine use EGR system it recirculates a small metered amount (6 to 13%) of the inert exhaust gas back into the intake manifold. The cooler exhaust gas absorbs heat from the much hotter combustion process. This reduces peak combustion temp^s & hence leaves the formation of NO_x . (NO_x is $\propto$ to peak comb. temp^s).

Controls of Emission from S.I. Engines (main are HC, CO, NO_x):-

There are two methods:-

- (1) To reduce the formation of pollutants by redesigning the engine system, fuel system, carburetor & ignition system.
- (2) By destroying the pollutants after these have been formed.

(A) After burner of Exhaust gas:-

(B) Catalytic converter

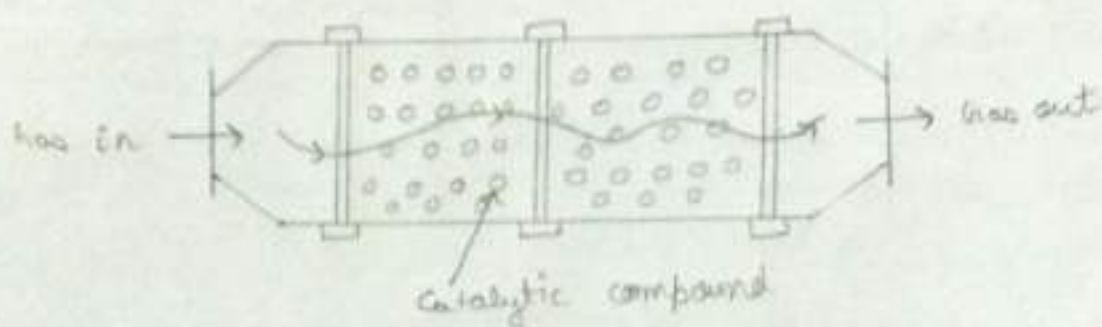


Fig:- Catalytic-converter

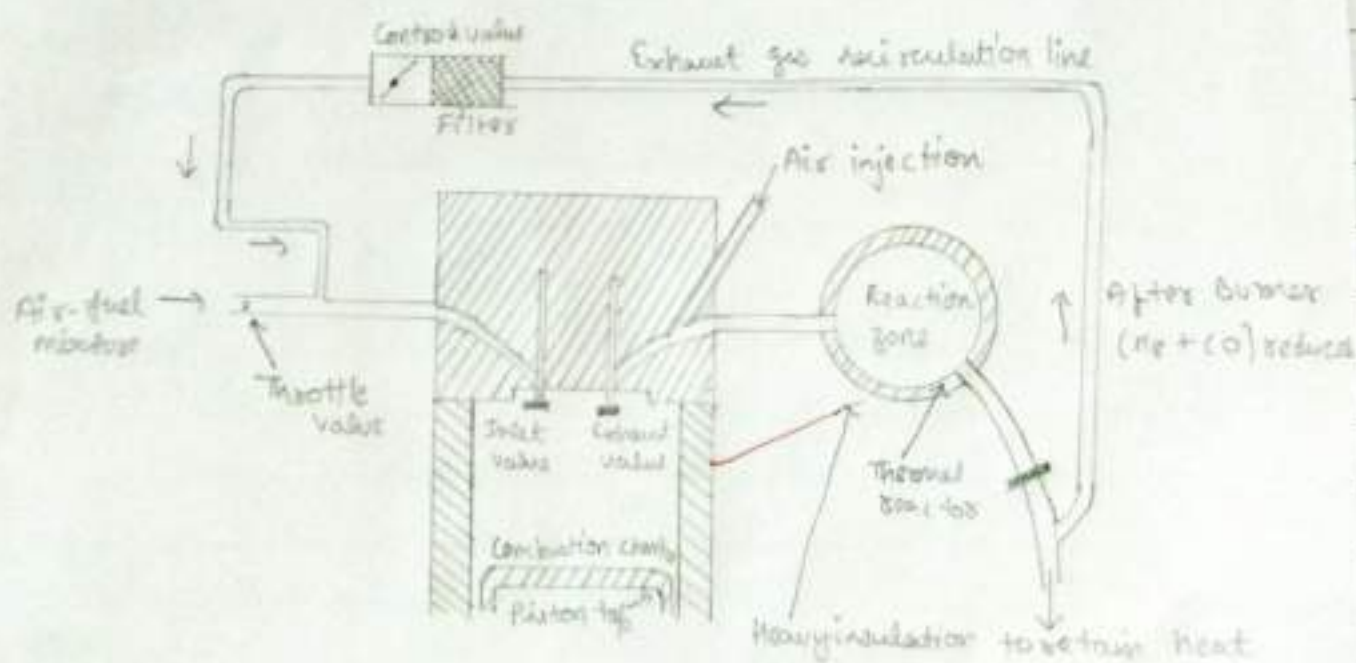


Fig:- EGR System



ENGCOL Con. Co-op. Stores Ltd., Jodhpur

M. B. M. UNIVERSITY JODHPUR

Date 4/5/23Name Zahid KhanExperiment No. 8Faculty No. 20/24765P. No. 61Class BE III Sem Batch A4

- By the oxidation of HC & CO is accomplished when placed in Exhaust system. (To reduce HC & NO).

(i) Al. oxide (Alumina) used on ceramic material for inside of converter (Can bear high temp^s) & chemically method.

(ii) Common catalytic material are

Platinum & Rhodium

&

Palladium

To oxidise HC & CO

To oxidise NO_x

⇒ Some data's of Euro6 :-

(i)

Petrol (g/kw)

	CO	HC	NO _x	PM	Sulphur
BS IV	1.00	.10	.08		30 PPM
BS V					
Euro6	1.00	.10	.06	0.005	10 PPM

To be shipped since 2018

Diesel (g/kw)

	CO	HC	NO _x	PM	Sulphur
BS IV	0.50	0.25	0.30	0.0025	50 PPM
BS V					
BS VI	0.50	0.06	0.17	0.005	10 PPM

To be shipped since 2018



M. B. M. UNIVERSITY JODHPUR

Date: 4/5/23

Experiment No. 9

P. No. 62

Name: Zahid Khan

Faculty No. 20/24765

Class DE II Sem Batch. A

* Object:- To study advanced valve timing system of a modern passenger car.

* Theory:-

→ Valves:-

(a) Intake valve is usually larger:- Force moving the mix into the cylinder in atm pressure. [Naturally aspirated - engines]

(b) Turbochargers & Superchargers } 35-60% more power can be produced than naturally aspirated engines.

(c) TC & SC (1) the pressure in suction manifold & more air or more A/F mix is forced into the cylinder which increases the engine power.

- Some more points on valves:-

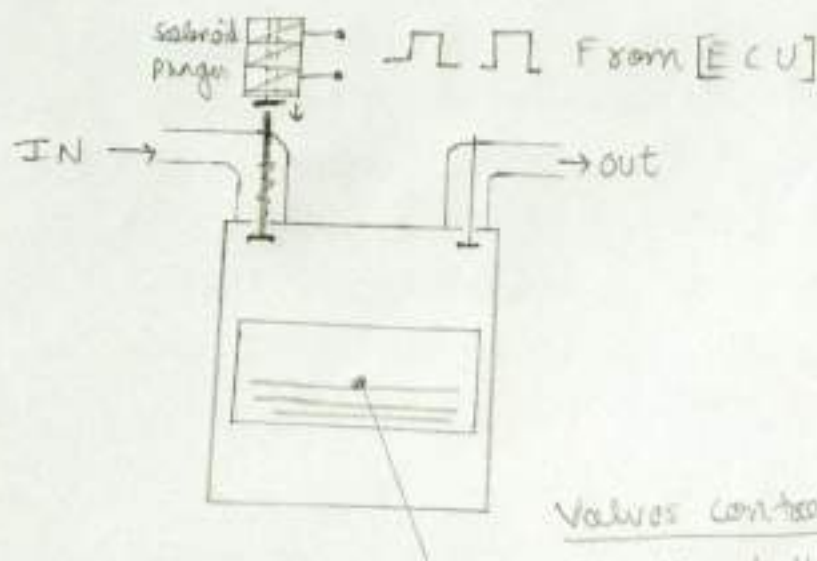
(a) IV is larger than Ex valve.

(b) Valve have hollow stem, reduces inertia & (response & power).

(c) Ex. valve may become red hot (871°C) when it seats on seat, it transfer heat to seat.



Valve timing diagram



Valves controlled
with the help of [ECU]



M. B. M. UNIVERSITY JODHPUR

Date 9/5/23

Experiment No. 9

P. No. 65

Topic

Name Zahid Khan

Faculty No. 20/24765

Class BE III Sem Batch. A4

* Variable Value timing:-

- Engine has lower volumetric η at higher speed because
- (i) short time to enter the A/F mix in cylinders on speed increases.
- (ii) If intake valve is opened earlier at high speed, more A/F mixture will start entering as it will start entering easier.

* Ways to open Intake valve earlier at high speed:-

- (i) By variable timing cam shaft.
- (ii) Electric solenoid open the valve.
- (a) Solenoid plunger is mounted against end of the valve
- (b) sensors send information on N_f , Load & other variables to ECM (Electronic control module). ECM finds how deep lmp & when the valve is to open. At proper time ECM sends a signal to solenoid. The plunger extends & open the valve. at higher speeds the ECM opens the valve earlier & may hold them open for long time.

* Variable value timing & lift:-

- Both valve timing & lift can be changed. This gives the engine smooth idle of a passenger car engine



M. B. M. UNIVERSITY JODHPUR

Lodha

Date 4/5/23

Name Zahid Khan

Experiment No. 9

P. No. 66

Faculty No. 20/24765

Class B.E. II Sem Batch Ay

4 the high speed performance of a racing engine.
The working is done by ECM as mentioned above.

→ Some Important note:-

	Petrol	Diesel
Compression ratio (γ)	6-10 (Avg 7-9)	12-12 (Avg 16-18)
Calorific Value	42 MJ/kg	Nearly same for petrol and
A/F ratio	10-17	18-200
BSFC	.3 kg/kWH	.27 kg/kWH
Not:-	waste of fuel	No waste
Specific gravity	.75	.84



M. B. M. UNIVERSITY JODHPUR

Date: 4/5/23

Experiment No. 10

P. No. 87

Name: Zahid Khan

Faculty No. 20/29/65

Class: BE III sem Batch: 1st

* Object:- To perform Morse test on Fiat petrol engine & Draw the performance curves.

* Theory:-

- Indicated Power (IP) is a theoretical power developed inside the engine cylinder during the combustion process. It represents the power generated by the fuel-air mixture, which drives the piston & rotates the crankshaft. Indicated power is usually expressed in units of horsepower (Hp) or kilowatts (kW).

- There are different methods to calculate Indicated power depending on the type of engine & the data available. Here are some common methods:-

(1) The william's line:-

- It is also known as the fuel rate extrapolation method. To determine the fictional power of an IC engine, a graph is drawn for brake power (kW) on the x-axis & fuel flow rate (kg/hr) on the y-axis on the condition that the engine is running at a constant speed.



M. B. M. UNIVERSITY JODHPUR

Date 4/5/23

Experiment No. 10

P. No. 68

Name Zahid Khan

Faculty No. 20/24 765

Class BE II sem

Batch A4

(2) The motoring test:-

- The test in which the engine runs at a constant speed using the motor & the engine is connected to the electric motor is called the motoring test.

(3) By Indicated diagram:-

- It is a chart used to measure the thermal performance of reciprocating steam & internal combustion engines & compressors. An It records the pressure in the cylinder versus the volume swept by the piston, throughout the two or four strokes of the piston which constitute the engine, or compressor, cycle.

(4) The Morse test:-

- The purpose of Morse test is to obtain the approximate Indicated power of multi-cylinder engine. It consists of running the engine against a dynamometer at a particular speed, cutting out the firing of each cylinder in turn & noting to fall in B.P. each time while maintaining the speed constant. When one cylinder is cut-off, power developed is reduced & speed of engine falls. Accordingly, the load on the dynamometer is adjusted so as to restore the engine speed. This



M. B. M. UNIVERSITY JODHPUR

ENGCOL Cam. Co-op. Stores Ltd., Jodhpur

Lodha

Date 4/5/23

Name Zahid Khan

Experiment No. 10

Faculty No. 20/21762

P. No. 69

Class B.E. VI Sem Batch 14

is done to maintain FP constant, which is considered independent of the load & ~~prop~~ proportional to the engine speed. The observed d/s in BP between all cylinders firing & with one cylinder cut-off is the IP of the cut off cylinder. Summation of IP of all the cylinder would then give the total IP of the engine under test.

* Formula:-

$$I = 4B - (B_1 + B_2 + B_3 + B_4)$$

where,

I = Indicated Power

B = Brake Power of all cylinders

B_1, B_2, B_3, B_4 = Brake power keeping that cylinder keep out.

Note:-

- 1) The number of cylinders is counted by the side of Jon.
- 2) The throttle value & the rpm remains the constant.



ENGCOL Cos. Co-op. Stores Ltd., Jodhpur

M. B. M. UNIVERSITY JODHPUR

Date: 4/5/23

Name: Zahid Khan

Experiment No. 10

Faculty No. 20/24765

P. No. 70

Class. BE 6 Sem Batch. A4

→ Some Questions to calculate B.P, I.P & mech η .

Q1

Ans-

If mech $\eta = 80\%$, & FP = 2kW, find B.P.

$$\eta = \frac{BP}{IP} \quad \& \quad IP = BP + FP$$

$$\therefore .8 = \frac{BP}{BP + 2} \quad \text{gives } \boxed{BP = 8\text{ kW}}$$

Q2

Ans-

If BP = 10, FP = 5kW, find I.P??

$$I.P = B.P + F.P = 10 + 5$$

$$\boxed{I.P \Rightarrow 15 \text{ kW}}$$

Q3

Ans-

~~Ans~~ If mech $\eta = 90\%$ & BP = 18, FP = ?

$$\eta = \frac{BP}{IP} \quad \& \quad IP = BP + FP$$

$$.9 = \frac{18}{18 + FP}$$

$$\therefore \boxed{FP = 2 \text{ kW}}$$



M. B. M. UNIVERSITY JODHPUR

Date: 4/5/23 Experiment No. 10 P. No. 71
 Name: Zahid Khan Faculty No. 20/24765 Class: AE II sem Batch: PY

→ Question on Morse test:-

~~Let 1500 RPM is~~

* Let engine is running at 1500 rpm keeping throttle value half open constant & the load of 10 kW.

Ans- B.P = 10 kW

$B_1 = 1 \text{ cut} \rightarrow$ then the load will decrease $\Rightarrow 7 \text{ kW}$

[Strong] $B_2 = 2 \text{ cut} \rightarrow \text{---} \Rightarrow 6.5 \text{ kW}$

[Weak] $B_3 = 3 \text{ cut} \rightarrow \text{---} \Rightarrow 7.5 \text{ kW}$

$B_4 = 4 \text{ cut} \rightarrow \text{---} \Rightarrow 7.0 \text{ kW}$

& the formula to cal. the indicated power is

$$I.P = 4 \times B - (B_1 + B_2 + B_3 + B_4)$$

$$I.P = 40 \text{ kW} - (7 + 6.5 + 7.5 + 7)$$

$$I.P = 40 - 28 = 12$$

21/11/23 55.23

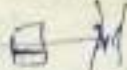
$$\eta = \frac{B.P}{I.P} = \frac{10}{12} = 0.833 \text{ or } 83.33\%$$

SP ₁	SP ₂	SP ₃	SP ₄
4	3	2	1

Number of cylinders ←
 Count from far side

MMT (3 stages)

2-Feb-23

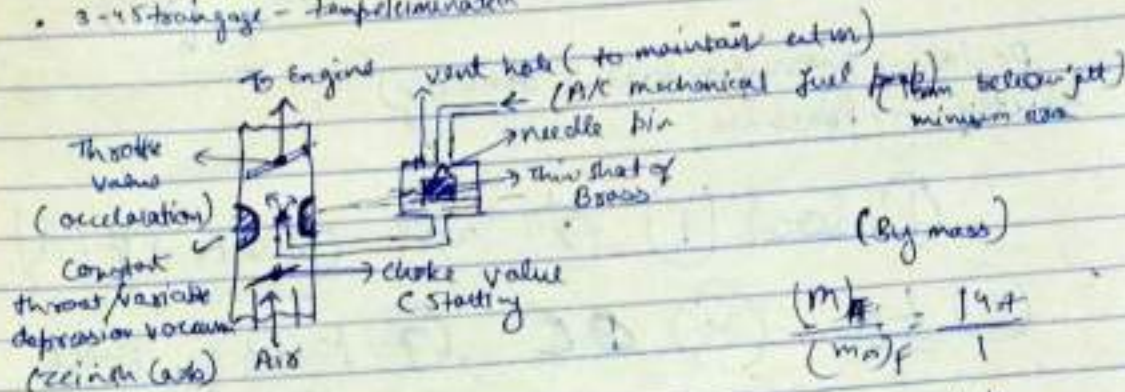


- 1) detector or sensor pickup or transducer (LVDT) , 2) Amplifier 3) Output

①

Zahid Khan

- 3-45 stage - tamper eliminated

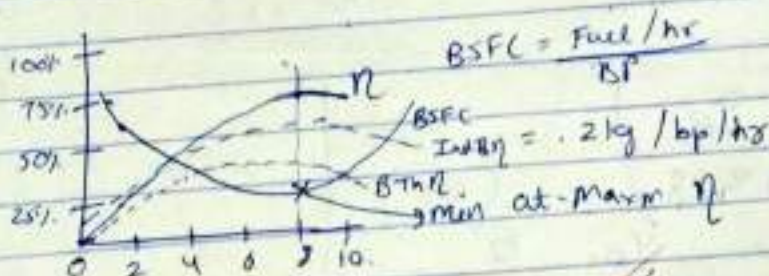


(By mass)

$$\frac{(m)_f}{(m)_f} = \frac{14.7}{1}$$

$$14.7 \text{ gm } \frac{9.8}{1 \text{ gm}}$$

Variable Choke — (5U)
& Constant depression



$$B.Th. \eta = \frac{E_p \text{ heat to BP}}{\text{Heat on Fuel } (C_v \times m_f)}$$

$$\text{Indicated th. } \eta = \frac{E_p \text{ heat to IP}}{\text{Heat on fuel } (C_v \times m_f)}$$

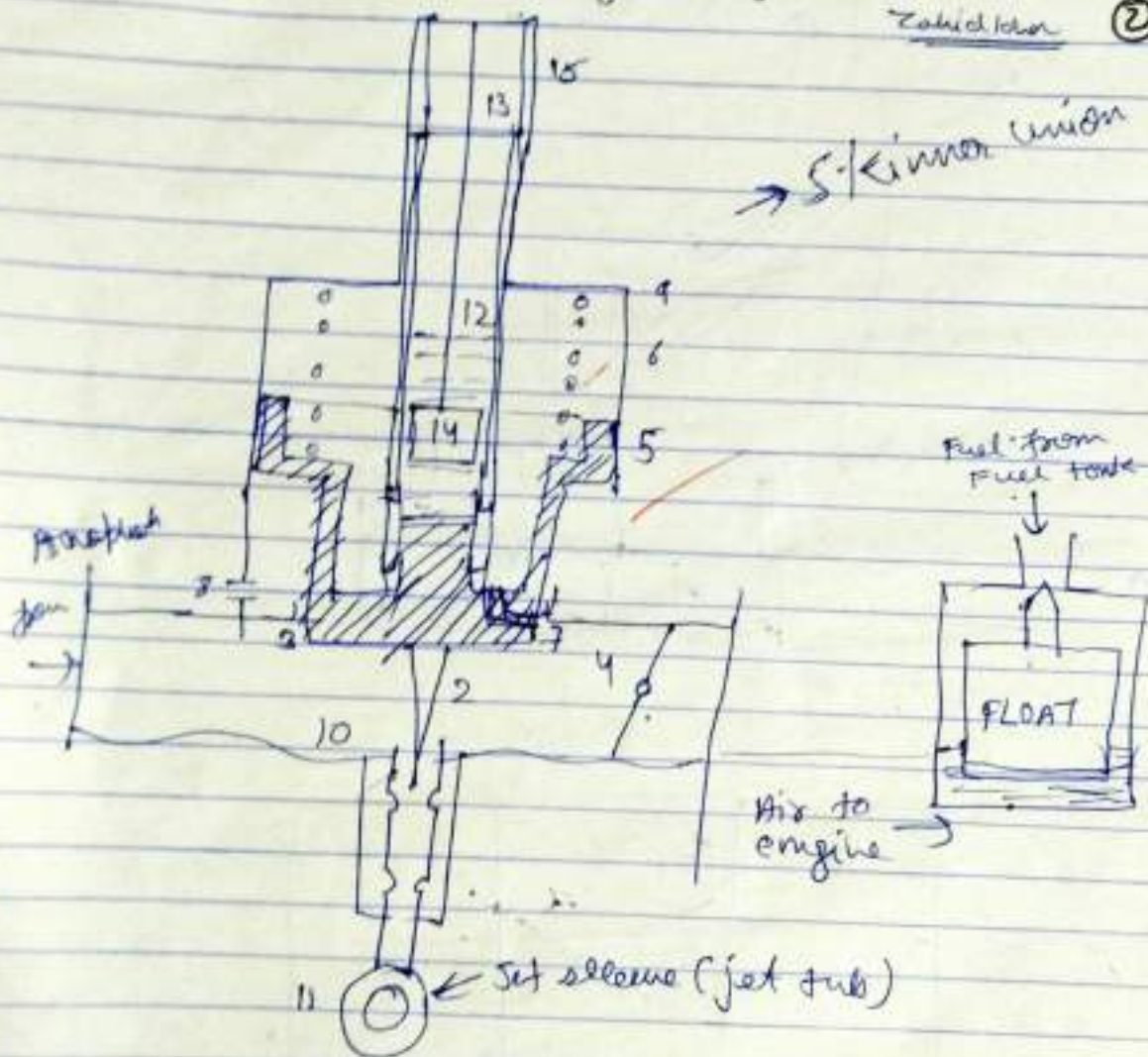
$$BTh \eta = \text{Ind th } \eta \times \text{mech } \eta$$

Zahid

* Objective: To study Various type of Carburetor
 (a) S.V. Carburetor (b) Zenith Carburetor

Schematic diagram of S.V. Carburetor

Labelled ②



11
2, 3, 23

- Experiment - To study
- apparatus required
- Theory ^{carburetor} _{S.V.}
- description (last working)
- (a) Zenith carb (fig)
- (b) S.V. carb (fig)

- Discussion (6-8 min)

⑪

Thermal lab

* Overall η = Brake η
(Not more than 40%)

9/3/23

Indication

③

Object :- To study Various cooling methods for IC Engines. (Air, water & coolant cooling)

→ Heat balance sheet (Heat expenditure)
(Calorific Value)

- ① Brake power
- ② Exhaust gases
- ③ cooling water
- ④ Radiation, conduction & unaccounted

Losses are nearly 30 to 35%

Indicated

Mechanical > thermal > Brake thermal η

* Friction should be as less as possible

* % of water Volume get increases on condensation to ice. (9 tak thaxi hogi toh 10 tak nojaygi)

pre-ignition - In only petrol engine (because of A/F mixture)
~~start~~ in diesel engine (Only air present)

* Aeroplane $\rightarrow$ 5/6 liter per sec

* Scooter $\rightarrow$.25 ml per sec

If cooling is not done

- ① Engine seizer (chipakna), distortion of engine block.
- ② Hot spot in the cylinder, pre ignition & loss of power
- ③ If engine temp is high then no doubt brake thermal η increases byt engine seizers take place.

(iv) Higher the cooling temp of water less is the friction because viscosity at high temp is lower. Zaid Khan (4)

(v) At high temp piston will expand considerably and seize the liner.

(vi) Excessive stress in parts rendering them useless for further operations.

(vii) At a high temp, lubricating oil will evaporate rapidly & piston & cylinder may damage.

(thermostatic switch)

Alcohol

• It goes from down space (Explain) its need to be filled completely & heating element cannot be covered properly & damage or distorted the system.

Ex- House heater (gases)

Chlorogens

(i) Air cooling (a)

(ii) Water cooling (b)

(iii) thermostatic switch (c)

(iv) Valve (d)

(70°C)

* desired characteristics of coolant.

(i) Low freezing temp, high boiling point

(ii) Large latent heat

(iii) Non corrosive

(v)

Easily & cheaply available

alcohol

• Role of anti freeze solⁿ: (Volume Expand)

- fracture of cylinder block, ^{water} jackets & pipes, radiator tubes may also be damaged.

Commonly used anti freezes & mixtures the additives are (i) Alcohol (ii) Glycerin

(iii) Kerosene

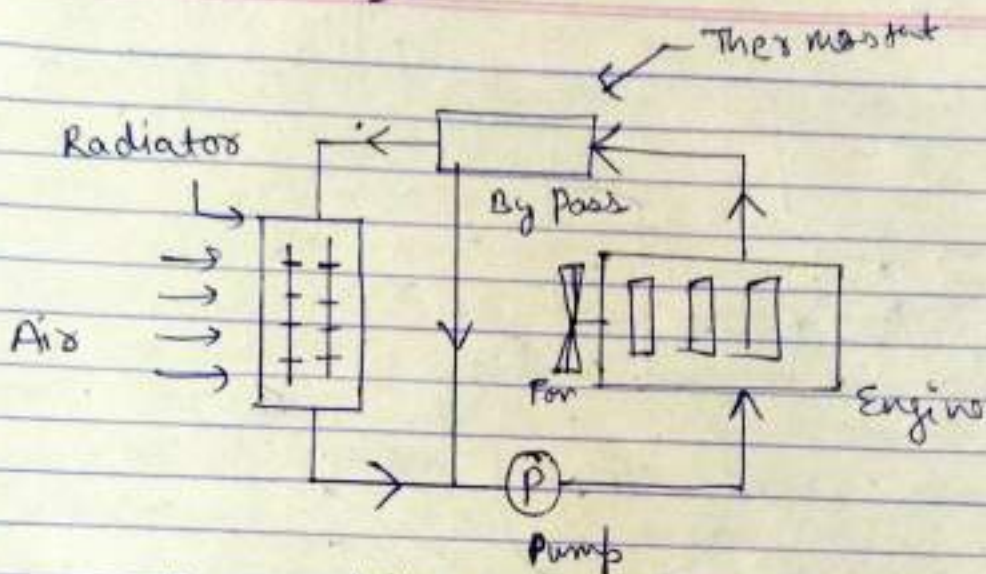
(iv) Sugar Solⁿ

(v) $CaCl_2$, $MgCl_2$

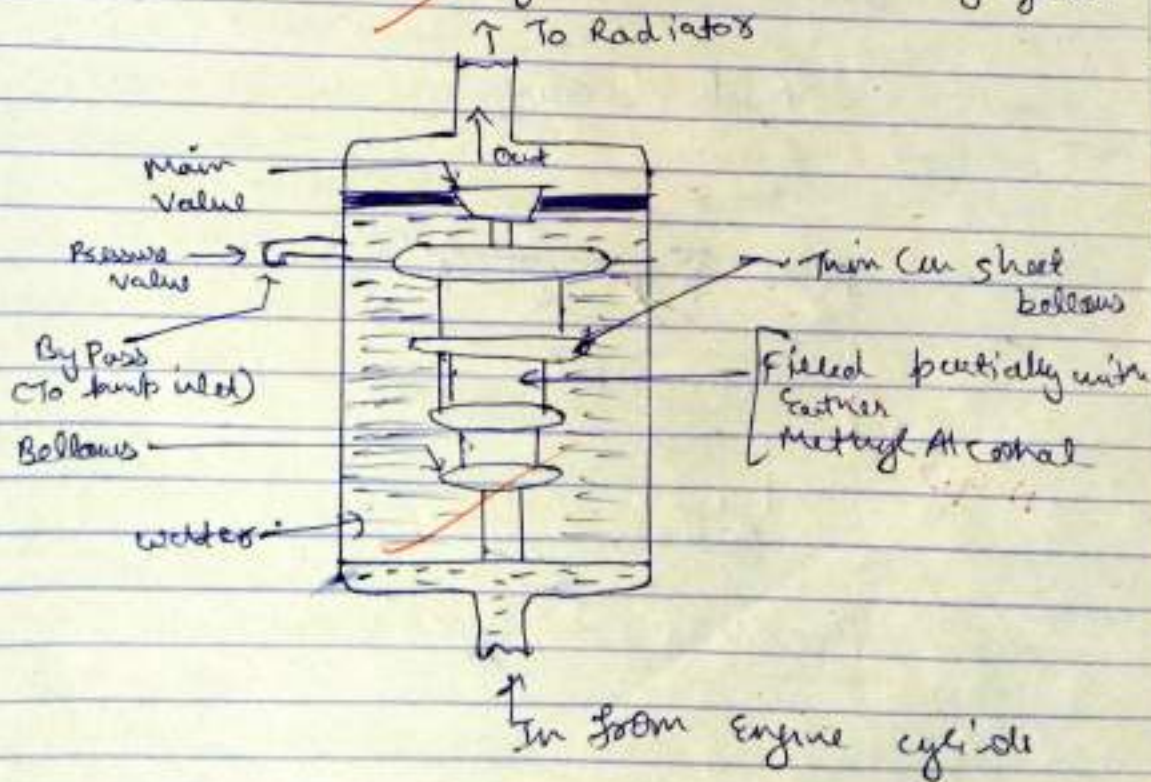
(vi) ethylene glycol (vii) propylene glycol.

Water Cooling

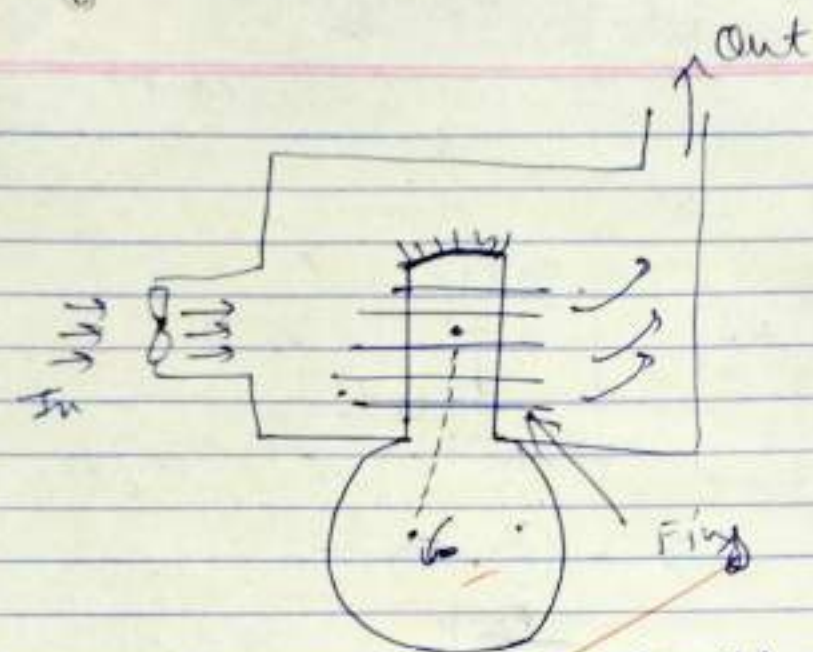
⑥



→ Thermostatically controlled cooling system



→ Typical thermostat



Small Engine & Aircraft

→ AIR Cooling

Pressure

↑



Vacuum
valve

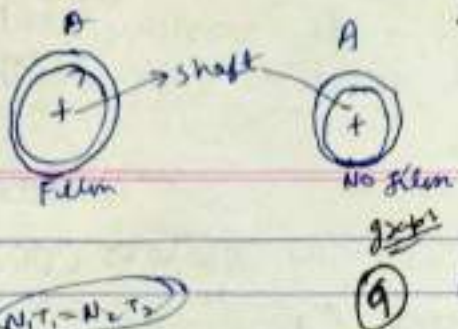


Radiator

Pressurized Cooling

4 diagrams

diagram 8



As $T \uparrow$ viscosity decreases which reduces friction loss upto a certain T . At T above $90^\circ C$ the local oil film destroys & metal to metal contact increases which causes more & more friction losses

Exp-4

$$N_1 T_1 = N_2 T_2$$

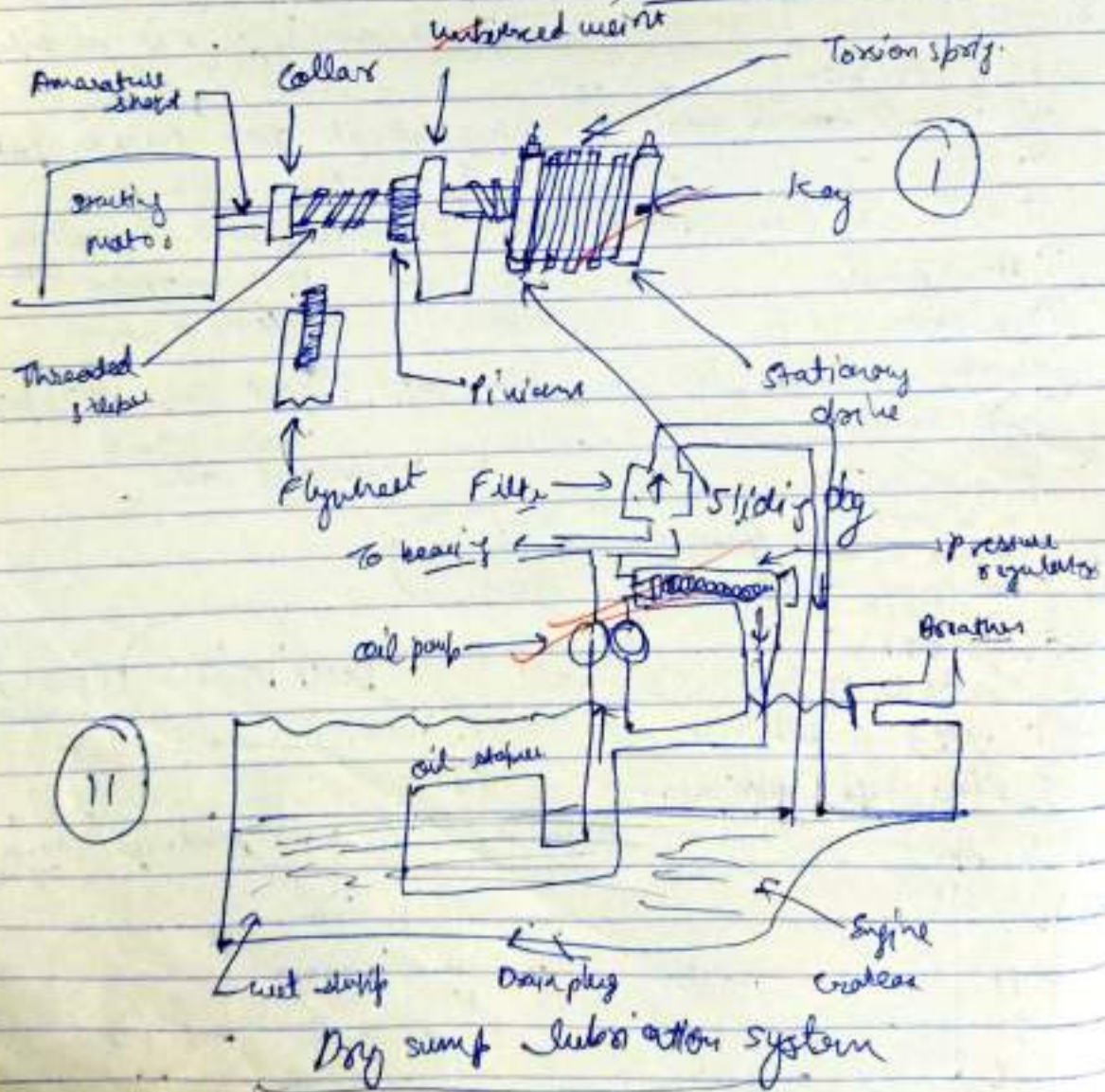
gear box $\rightarrow$ high torque shaft (propeller shaft & differential)

- workshop $\rightarrow$ 3 H.P. or 1 Mega watt
- $\rightarrow$ An very fast
- $\rightarrow$ Electric power to work

direct engine (battery $\uparrow$) requires more torque.

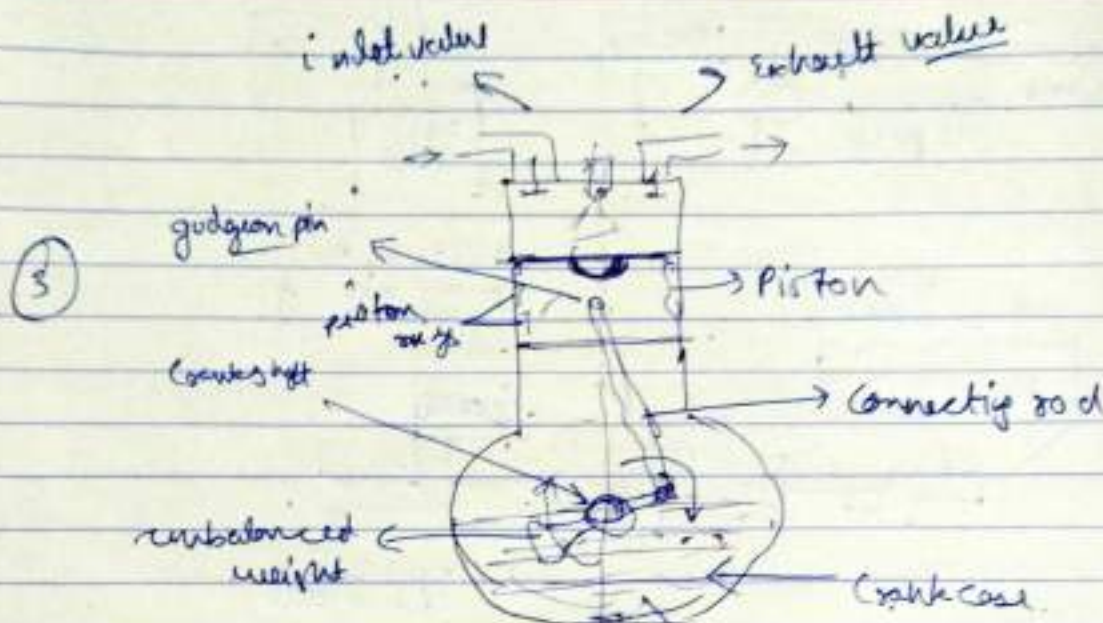
5:5
 $\downarrow$
strength to outside

Manual Method, Compressed air, Electrical method
Bendix drive



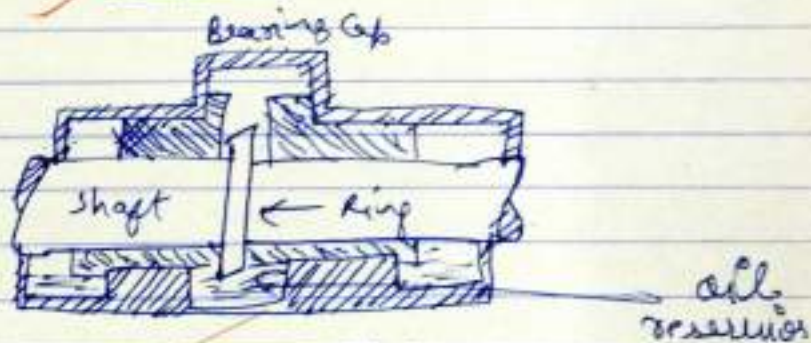
(11)

(10)

Zahid Khan

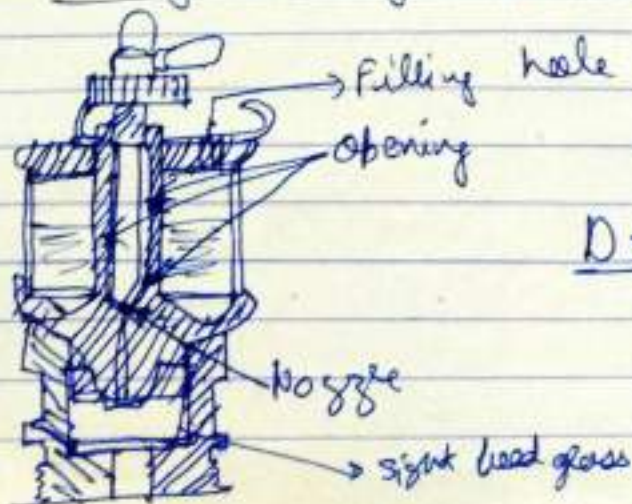
* Splash lubrication system

(4)



* Ring oiling system

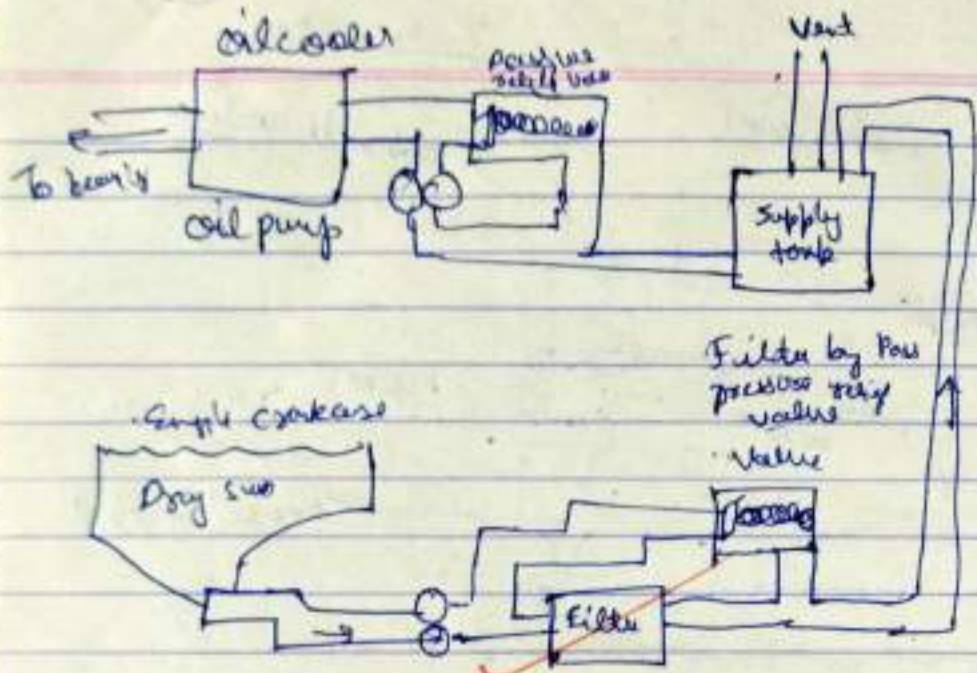
(5)



Drop feed oil cup

(1)

(11)



D 5 LG

11/6/4/25

Exp-5 To study DC generator & alternator as storage battery charging device. (AC)

Exp-6 To study advanced ignition system for SI engines. (Petrol engines)

13/4/23
(5)

* -ve wire bond to thick wire in vehicles

12 Volt system
(Now a days)

[required thick wire
overload less voltage]

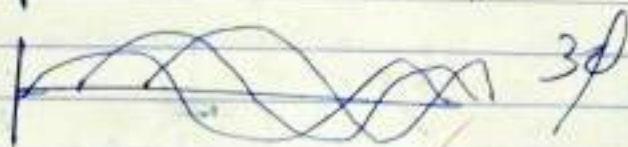
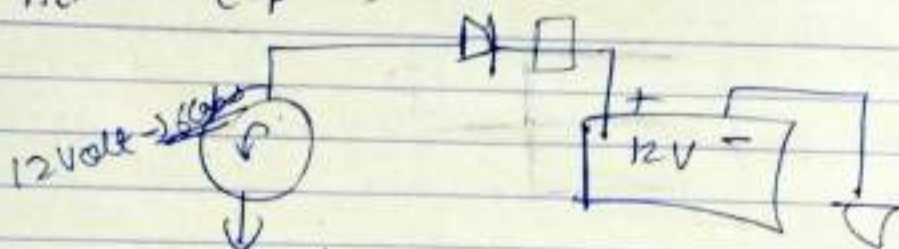
6 Volt system

2 horse power = 1500 Watts

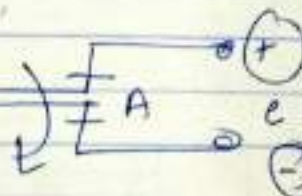
→ Amps capacity

charge $\Rightarrow$ (Amps/10) Not more than that

Alternator (3 phase)



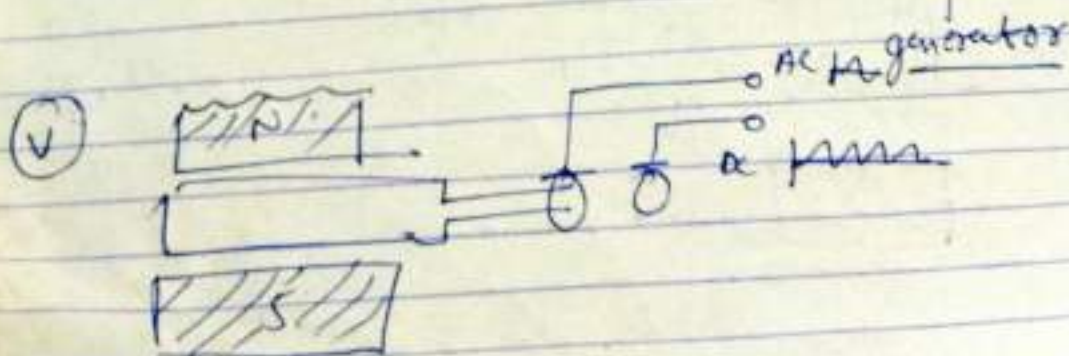
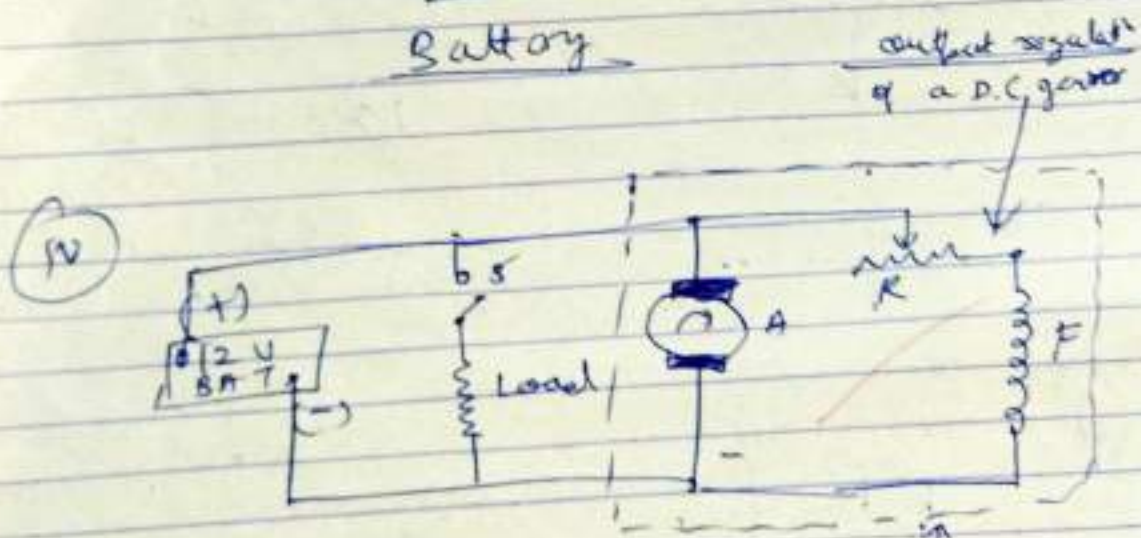
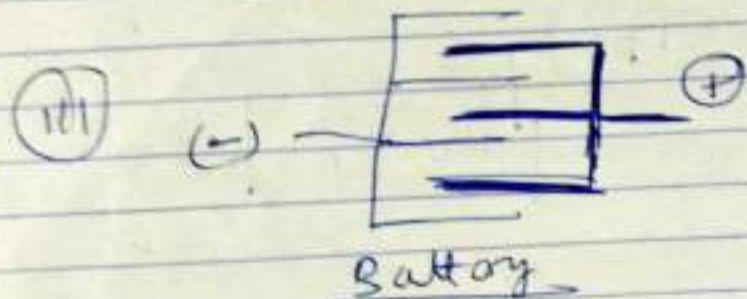
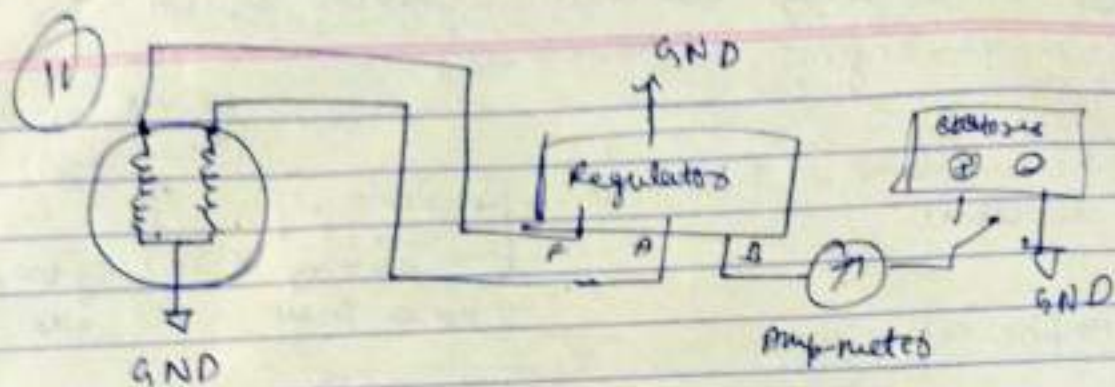
(1)



$$e = BLV \times 10^{-8} \text{ volts}$$

(A)

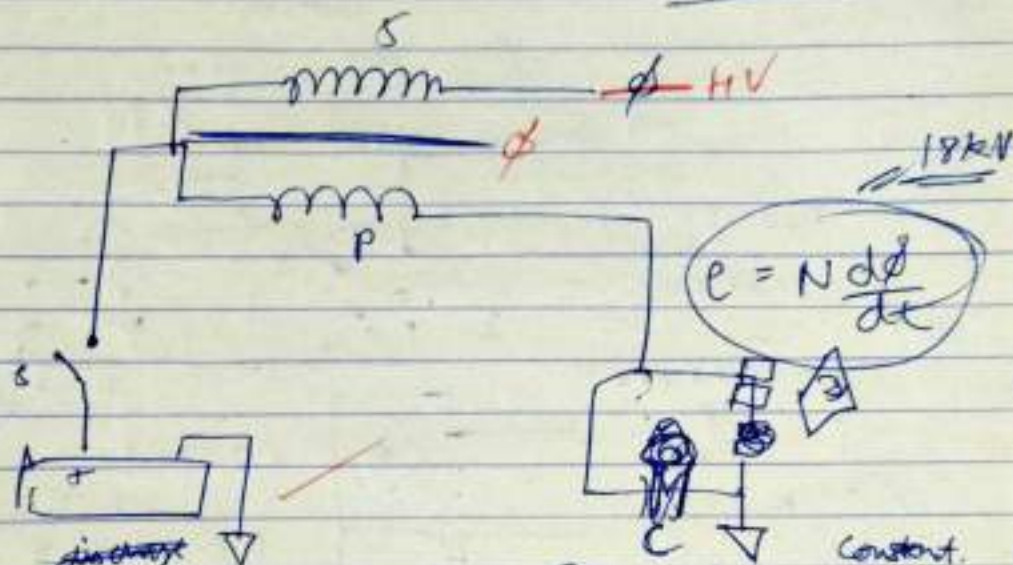
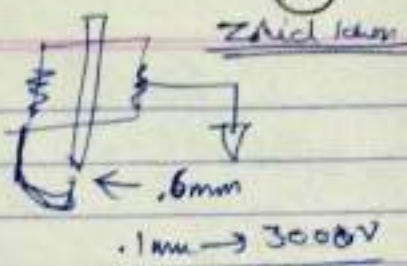
(B)



⑥ 1-3-4-2 (Balancing, 4 power generated from all engine (1, 3-2-4) also useful)

for a gap of point 16 mm
 $\downarrow$
18000V

metal blade $\rightarrow$ make voltage



① $\rightarrow$ coil ignition system (CIS)

② Capacitor discharge system (CDS)

③ transistor assisted ignition system (TAIS)

④ Piezoelectric ignition system (PEIS)

Single cylinder 4 stroke engine, running at 3000 rpm
 spark advanced 30°, sec in one

$$\frac{360}{480} \times 30 = 3000 \text{ rpm}$$

$$5000 \text{ V/sec}$$

$$50 \times 360^\circ / \text{sec}$$

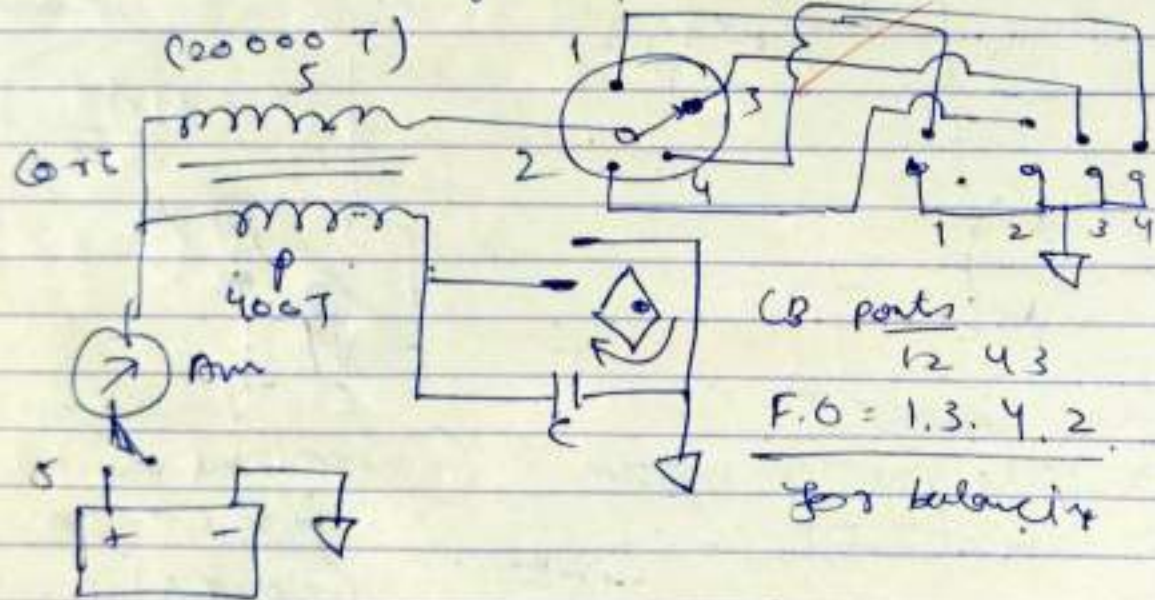
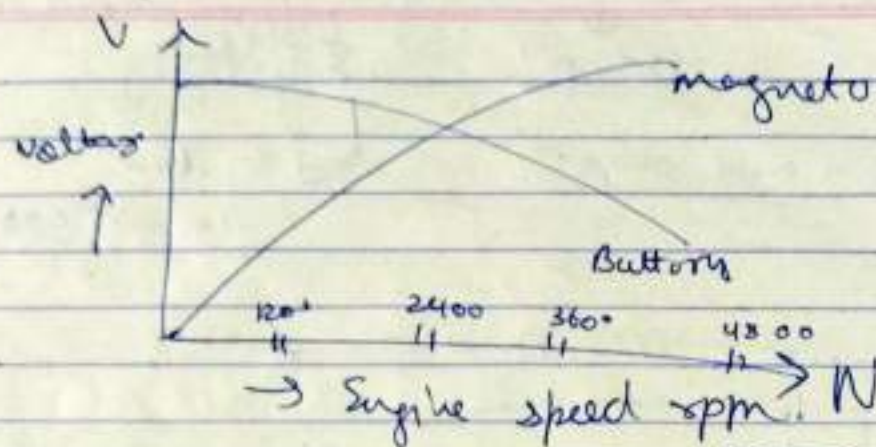
$$360^\circ \rightarrow \frac{30}{50 \times 360} \rightarrow 0.00167 \text{ sec}$$

$$= 1.67 \text{ ms}$$

→ New only air is taken as input

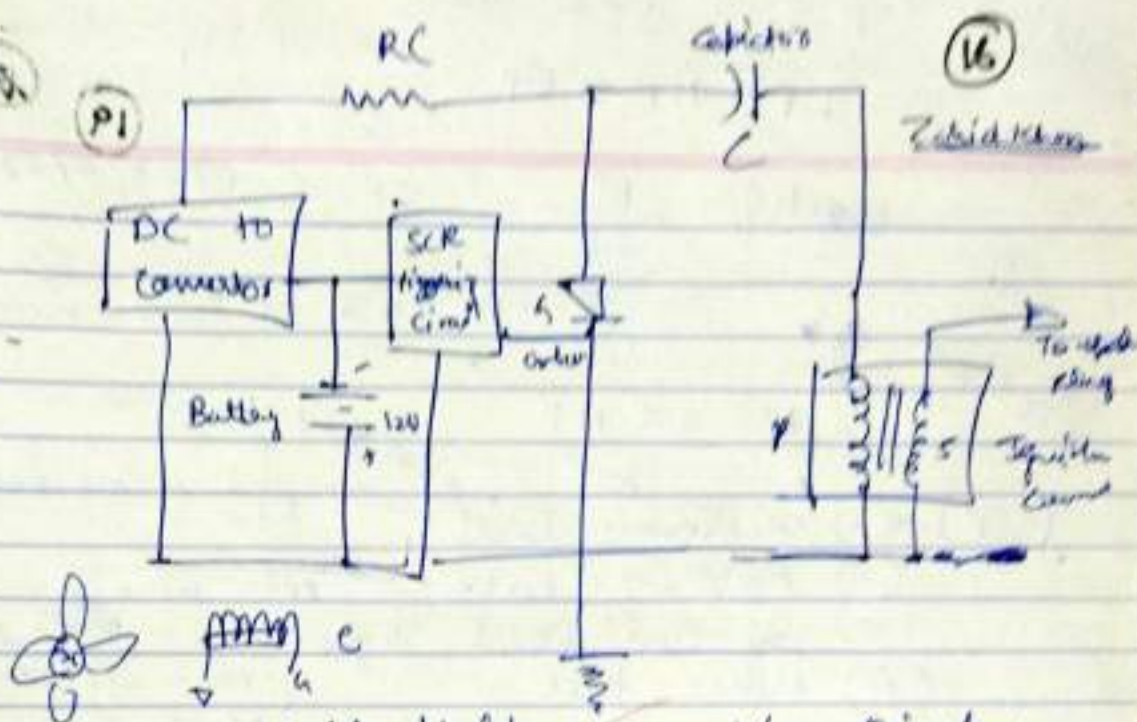
(17)

(B)

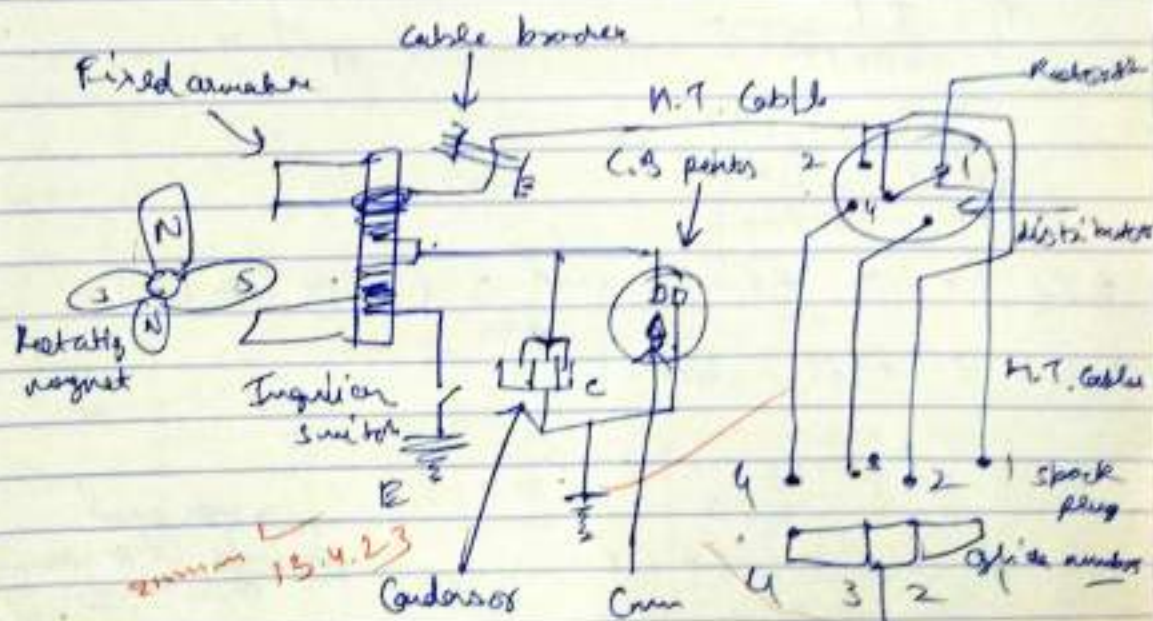


SP gap . 45 mm - . 85 mm

Ignition system [coil & magneto]



→ High Voltage Capacitive Discharge Ignition System (COIS)



Layout of magneto ignition system. Various parts 1-3-4-2 on distributor indicate firing order.

4/5/23
21

Exp-10 → Morse test

$$IP = BP + FP$$

Exp-10
19

$$\text{mech } \eta = \frac{BP}{IP}$$

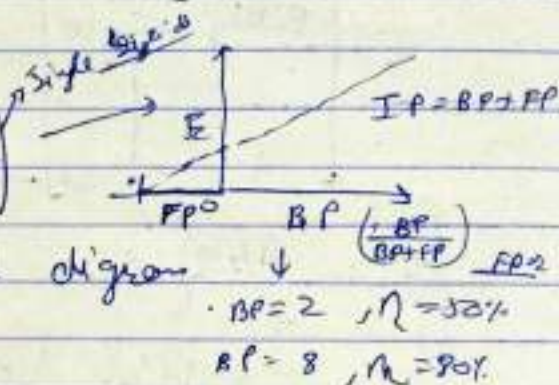
$$IP = 10, BP = 8, FP = 2$$

$$= \frac{8}{10} \Rightarrow 80\% \text{ or } 0.8$$

W_h 2000
→ dyno

$$BP = 2 \pi NT$$

- (IP) → (i) William's line
(ii) Motoring test
(iii) By indicated diagram
(iv) Morse test



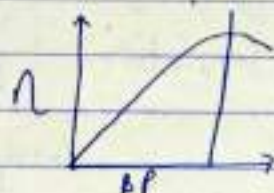
Q If η is 80%, $FP = 2$ find

$$\eta = \frac{B.P.}{B.P. + F.P.} = 0.8$$

$$B.P. = (0.8)(B.P. + 2)$$

$$12 B.P. = 168$$

$$B.P. = 14$$



Q If $BP = 10, FP = 5$, find $IP \Rightarrow 15 (BP + FP)$

Q Mech $\eta = 90\%, BP = 18, F.P.$

$$0.9 = \frac{18}{18 + F.P.}$$

$$FP = 2$$

$$16.2 + 0.9 FP = 18$$

Thick wire used
terminals (more power)
diesel engine
(less compression ratio)

→ motor's charac waki BP nikala, Refused engine

(10) indicated dly



$$IP = \frac{P_m L A N}{K}$$

* Object:- To study advanced fuel injection system (Electronic control module) petrol
5 Multipoint fuel injection system (MPFI), Common rail direct injection system (CRDI). -diesel
20/4/23

Q Why we have rejected carburetor?

1. Carburetor fails to maintain air-fuel ratio at all conditions.
2. There is resistance to flow of mixture by throttle valve.
3. Venturi also obstructs
4. Since we use supercharger & turbocharger. (turbo-charger exhaust gas) (engine output)

(17)
Zahid Khan

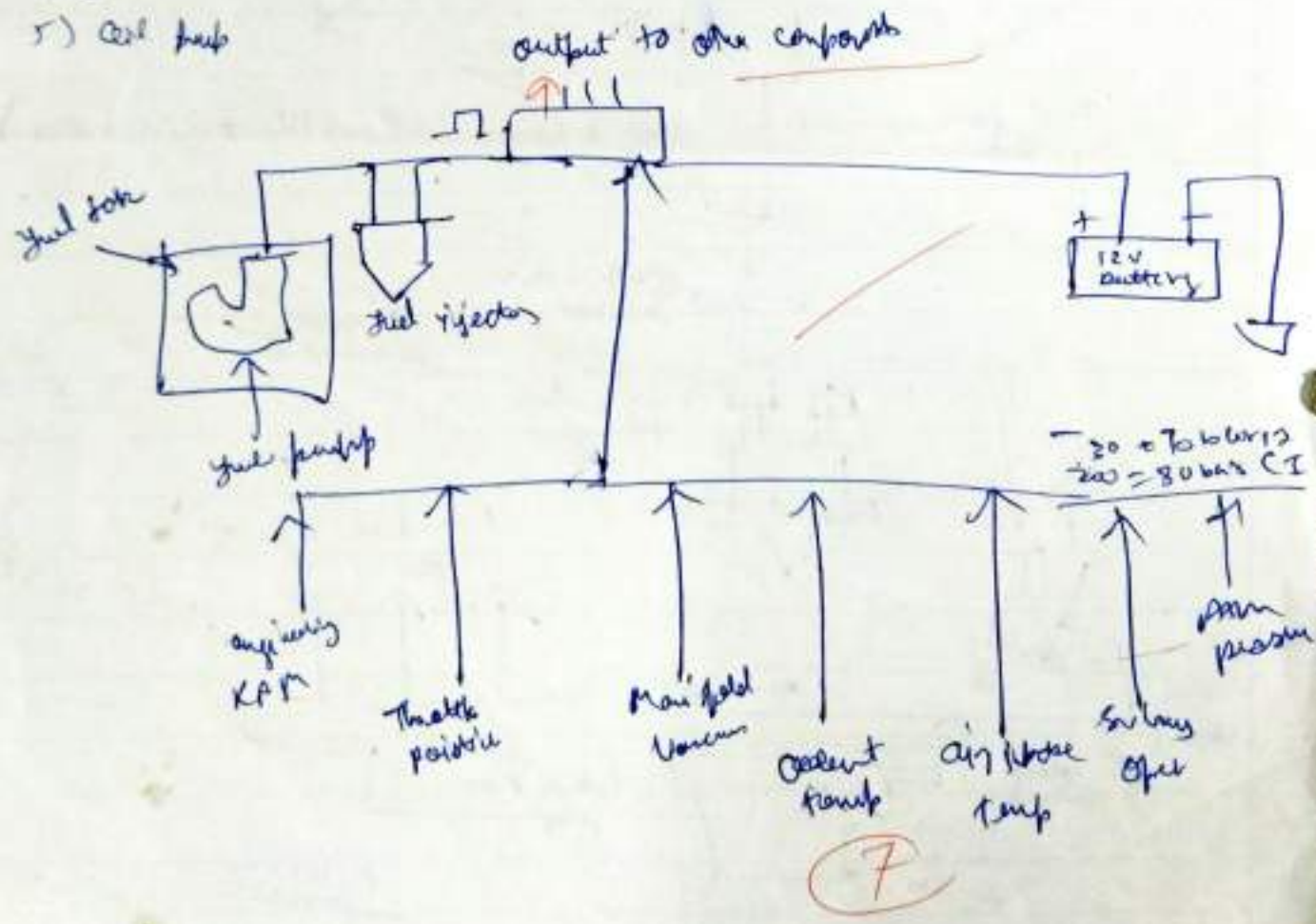
Merits of injection system

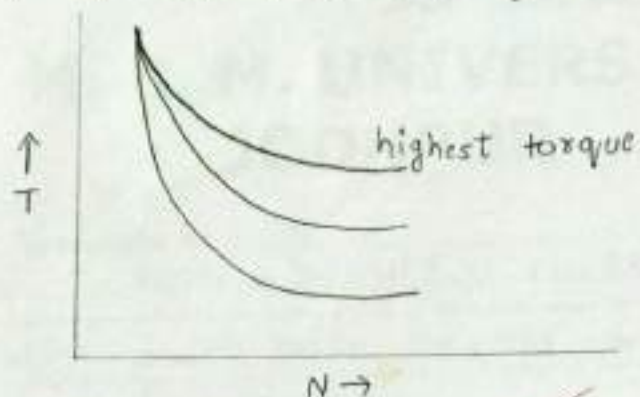
- a) High Volumetric η as no venturi restriction. (Air intake T)
- b) Jet for preheating of air fuel mixture is added.
- c) Acceleration response is better
- d) Atomization is greatly improved

Cut section of CRDI engine (4 cylinders 4 stroke)

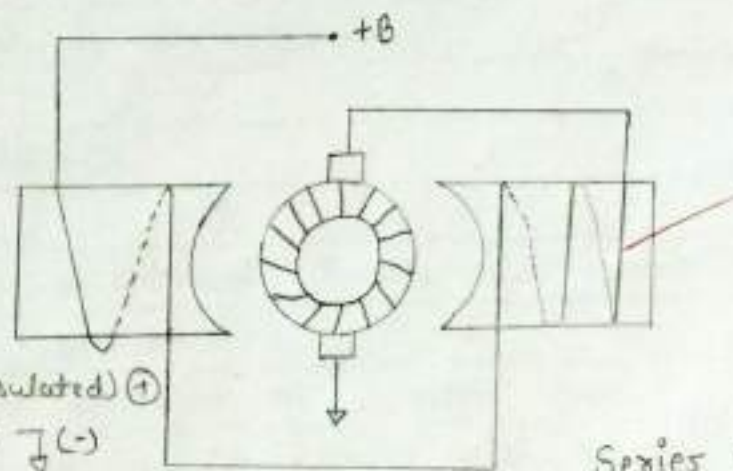
- 1) water pump
- 2) engine block
- 3) cam
- 4) common rail
- 5) fuel pump

petrol engine



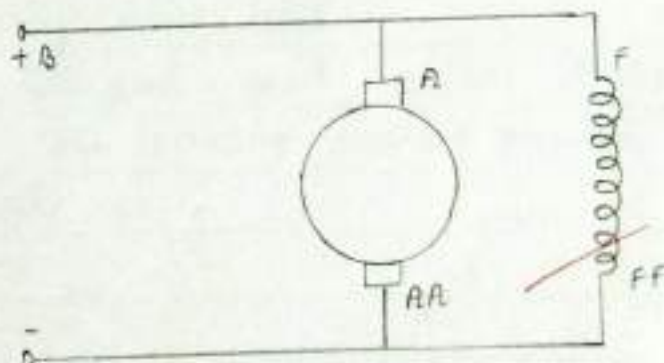


Relation between motor speed & torque



- Motor terminal (insulated) $\oplus$
- Other is grounded $\ominus$

Series Motor



Shunt Motor



M. B. M. UNIVERSITY JODHPUR

Date 6/4/23Experiment No. 4P. No. 31Name Zahid KhanFaculty No. 20/29765Class BE II Sem Batch A1

3. The ignition switch on a solenoid such which sends this high current in self starts.
4. In house Inverter of 50W/12V the ovp is 70 Amp. which is thought a thyristor or SCR.

(b) Bendix-drive :-

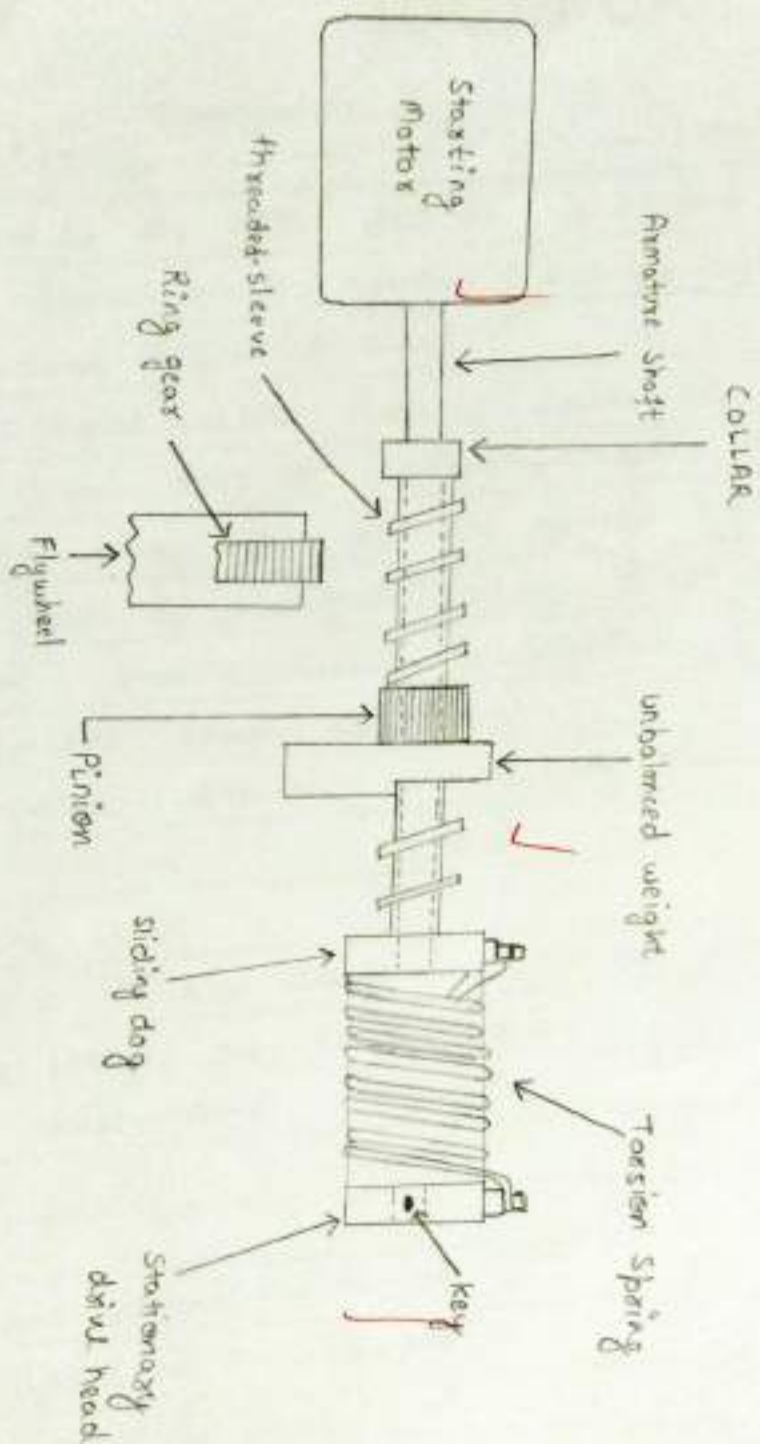
- Construction:- A standard Bendix drive consists of the following main parts:-

- (i) A threaded sleeve on the armature shaft.
 - (ii) An unbalanced weight attached with the heavy pinion.
 - (iii) A stationary drive head keyed on the armature shaft.
 - (iv) A sliding dog fastened over the threaded sleeve.
 - (v) Torsional (helical) spring attached at its two ends to the sliding dog & the stationary drive head.
 - (vi) A collar as an integral part of the threaded sleeve.
- And off-course the starting motor & flywheel with the ring gear are essential.

• Working:-

- As soon as the motor is started, the armature shaft rotates but the pinion mounted on this shaft does not rotate with it because of the unbalanced weight. It, therefore, moves axi-

Fig: Details of a standard Bendix drive having a torsional spring.
The mechanism is 'Inboard Type'.





M. B. M. UNIVERSITY JODHPUR

Date 6/4/23

Experiment No. 4

P. No. 33

Name Zahid Ikhon

Faculty No. 20/24765

Class BE II sem Batch Ay

- ally towards the motor till it is stopped by the collar.
Now the pinion starts rotating & engages with the flywheel ring to cause engine to start.

- Once the engine start running under its own at a speed greater than that of a starting motor, the pinion is spun back along the screwed spindle & goes out of engagement. The function of torsional spring is to bear the shock of engagement thereby causing a smoother start.

- Some important points are:-

- (1) Pinion meshes with ring gear & comes back when engine is started.
- (2) The ratio is 1:15.
- (3) Bendix drive is also known as inertia drive.
- (4) It is also known as inertia drive.
- (5) It is a series motor.

• Note:-

- Motor rpm is 1000-2000 rpm
- Cranking speed is 80-120 rpm
- Reduction is 15-20 times
- The engine rpm is say 4000 rpm
- * If after starting motor the pinion does not disengage then motor will run at $4000 \times 15 = 60,000$ rpm & will damage the



M. B. M. UNIVERSITY JODHPUR

Date 6/4/23

Experiment No. 4

P. No. 34

Name Zahid Khan

Faculty No. 20/24765

Class DE II Sem Batch Ay

motor.

* Discussion :-

- The engine starting motor, which is typically a DC motor, is responsible for starting an internal combustion engine. The Bendix drive is a gear mechanism that helps engage the starter motor with the engine's flywheel to start the engine. Together, the starting motor & bendix drive ensures the engine is started smoothly & quickly.
- * There are two more types of arrangement systems are manual, method, compressed air method
(Also Bendix method is electrical method)

11/18.4.23



M. B. M. UNIVERSITY JODHPUR

Date 13/04/23

Experiment No. 5

P. No. 35

Name Zahid Khan

Faculty No. 20/24765

Class BE IT Sem Batch 14

* Object:- To study DC generator as storage battery charging device.

* Theory:-

① DC generator as Battery charger:-

A DC generator works on the principle of electromagnetic induction. It consists of stator, which is stationary part. It houses the field windings & a rotor, which rotates & houses the armature windings.

When the rotor rotates, it cuts through the magnetic field produced by the stator's field windings, generating an electromotive force (EMF) in the armature windings. This (EMF) can be used to charge a storage battery.

It is connected to a battery through protecting circuit like relay diode, etc. There is also a voltage regulating which inserts field resistance or cuts it back.

• Types of battery:-

① Lead acid batteries:- These batteries are widely used in



M. B. M. UNIVERSITY JODHPUR

Date 13/4/23

Experiment No. 5

P. No. 26

Name Zahid Khan

Faculty No. 20/24765

Class AE III Sem

Batch An

Cars, boats & Backup power system.

② Lithium-ion Batteries :- These are the most type of Rechargeable battery batteries used in the market for portable & electronic devices. such as smartphone & laptops.

③ Nickel - cadmium batteries :- These batteries are commonly used in cordless power tools, Toys & medical equipments.

④ Zinc - carbon batteries :- These batteries are similar to alkaline batteries but are less expensive & have a shorter life span.

⑤ Alkaline batteries :- These are the most type of disposable batteries used in household devices such as remote control & flash lights.

⑥ Nickel - metal hydride Batteries :- These batteries are often used in hybrid cars, digital cameras & portable electronic devices.

• Construction of battery :-

(1) Electrode

(3) Separator

(2) Electrolyte

(4) Containers

- (1) Rating in AH - 60AH — 1A - For 60HR.

(2) Charging ration = $\frac{AH}{10}$ Amp ; 240 H for 15min

(3) Full charge

Discharge

(Test by Hydrometer)

specific gravity = 1.280

specific gravity (1.175)



M. B. M. UNIVERSITY JODHPUR

Date 13/4/23

Experiment No. 5

P. No. 37

Name Zahid Khan

Faculty No. 20/29765

Class III Sem

Batch AM

(1) Separator - Lead sulphate formation (standard in discharge conditions)

(2) Alternator (A.C. generator):-

Some cars & vehicles require more electricity (Power windows, elect-equipments for Automatic transmissions) often vehicle run at low speeds in large cities & in heavy traffic 100 Amp to other so many advantages alternative are widely used:-

(i) AC - DC by diode (cheaply available)

(ii) No-cutout is required

(iii) Much smaller size

(iv) light in weight

(v) At idling speed also charges batteries

(vi) Less maintenance, more reliable.

• Construction:- In small magnet can be selectable keeping coil fixed. In for slip rings are used to collect current.

- Regulations:- Used to regulate the output of Alternator by limiting current in the field by electromagnet cuts.

~ Batt



M. B. M. UNIVERSITY JODHPUR

Lodha

Date 13/4/23

Experiment No. 5

P. No. 28

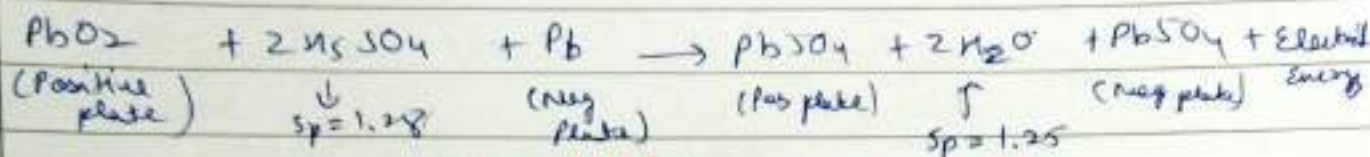
Name Zahid Khan

Faculty No. 20/24765

Class BFT Sem

Batch A4

Battery charging & discharging :-



Shows direction of CHG & DISCHARGING

- (i) Neg plate are more than $\oplus$ plates, so that action occurs on both the sides of the plates.
- (ii) When charged $\oplus$ plates are grey in colour.
- (iii) Separators - wood, rubber, orlon, fiber etc PVC
- (iv) Containers - hard glass or hard rubber.

Note:- Check always drive belt or fan belt for low charging & cooling problems.

Cutout relay :-

The cutout relay closes the circuit b/w the generator & the battery when generator is producing the current. Also, it opens the circuit so that the battery cannot discharge back through the generator when the generator stops or slows down.

Fig shows the schematic arrangement of cut-out relay it is simply a magnetic switch it consists of 2 windings, shunt winding & series winding, assemble

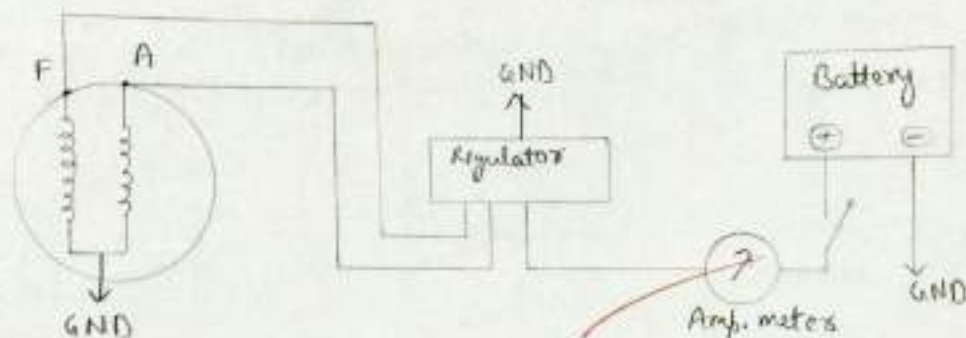


DC generator

Page - 39
salut 1 km

$$E = BLV \times 10^{-8} \text{ Volts}$$

- (2) Armature
- (3) Commutator
- (4) Brushes
- (1) Field



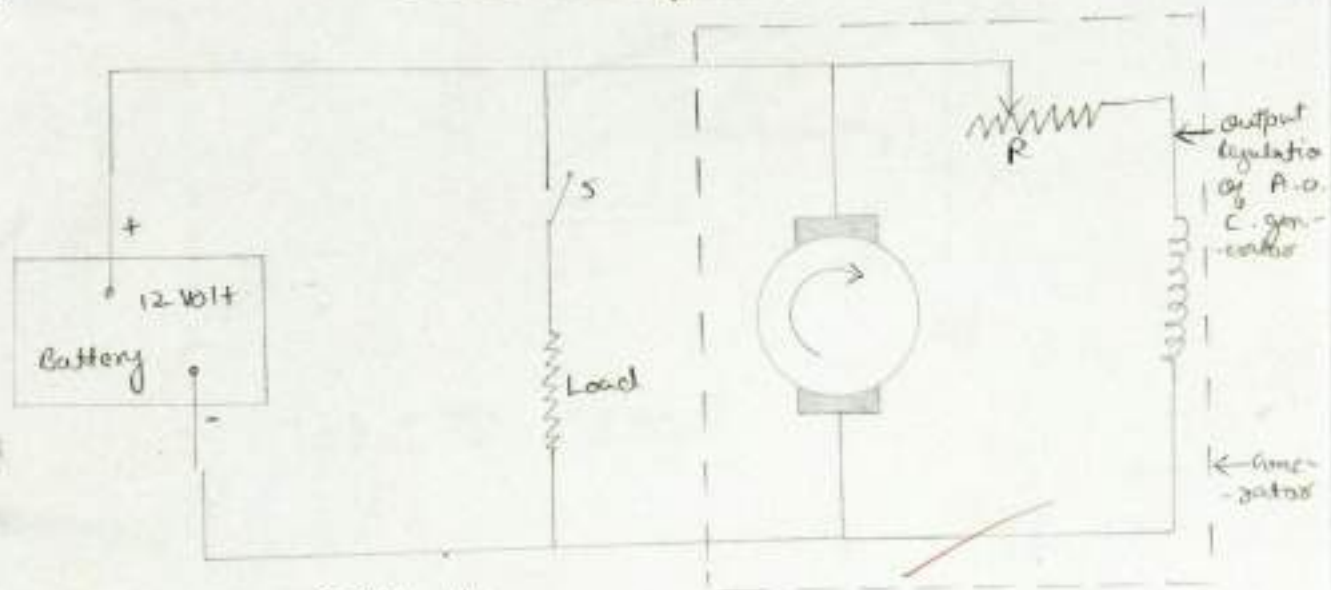
CHARGING-System

For 6V BAT =

6.75 - 7.5 volts

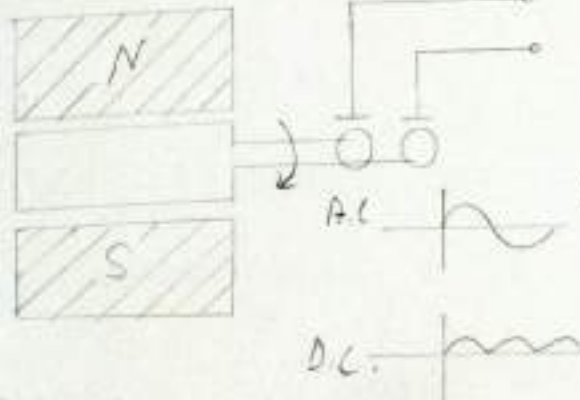
For 12V Bat. =

13.4 - 14 volts



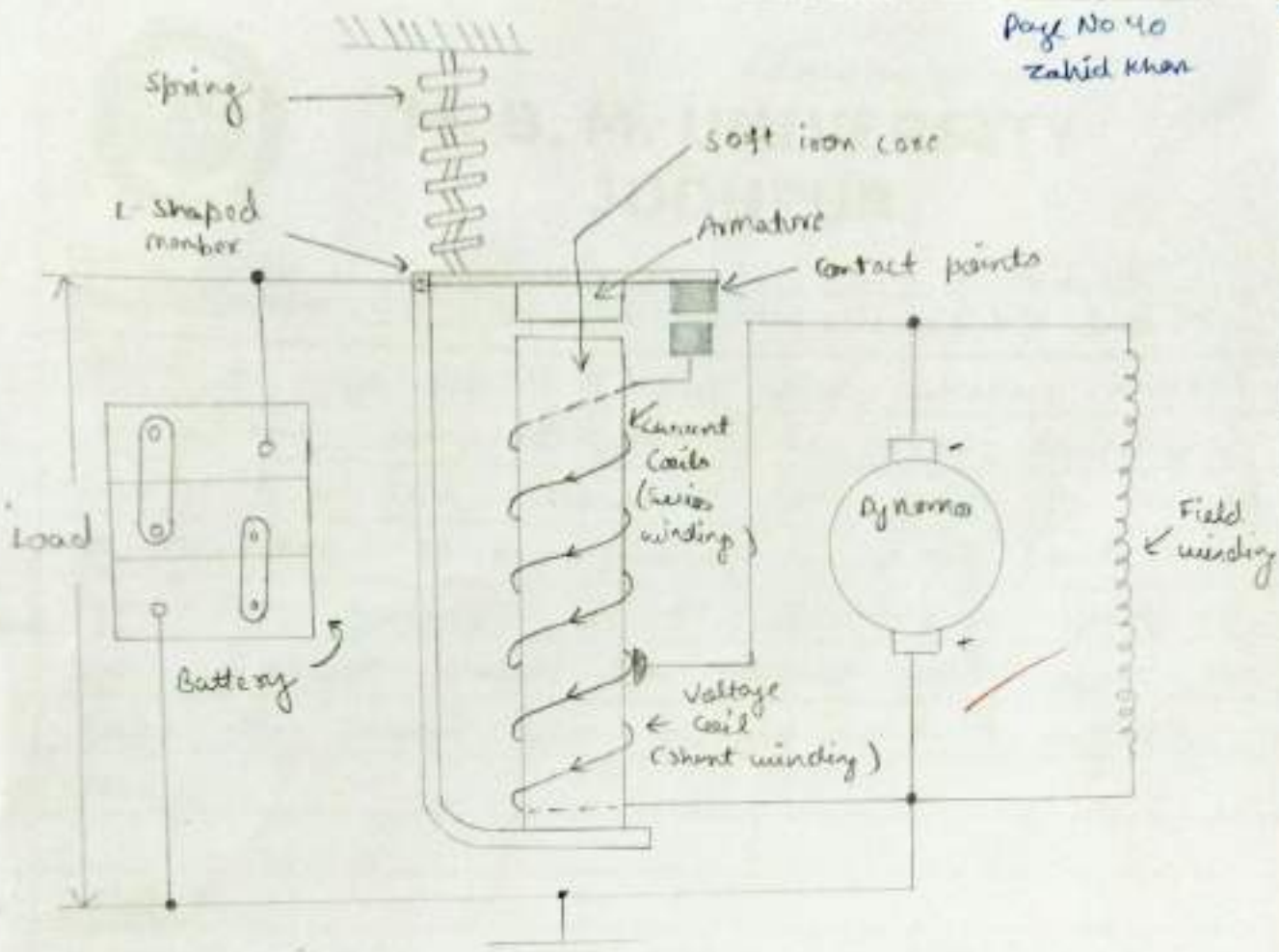
Sulphation

A.C. Generator



Battery charging & discharging





Cut - Out Relay



M. B. M. UNIVERSITY JODHPUR

Date 13/4/23

Lodha

Name Zahid Khan

Experiment No. 5

P. No. 41

Faculty No. 20/24765

Class 6 Sem

Batch A4

mount a soft iron & a flat steel armature mounted on a hinge above the core. The series windings consist of a few turns of fine wire. The armature is connected to the battery by contact points. When the generator is not operating, a spring on the armature holds the contact point apart, this keeps the circuit b/w the generator & battery open.

* Discussion :-

Overall, studying DC generators as Battery charging devices can you understand the principle of electromagnetic induction & the practical application of DC generator in power generation & storage.

11/4/23



M. B. M. UNIVERSITY JODHPUR

Date 13/04/23

Experiment No. 6

P. No. 42

Name Zahid Khan

Faculty No. 20/24765

Class III Sem

Batch PM

* Objective :- To study advanced ignition system for SI (Spark Ignition system) engines.

* Apparatus required :- Magneto ignition system, Electronic Ignition, Transistorised assisted contact ignition system (TACIS), High voltage capacitive discharge ignition system (CDIS).

* Theory :-

- Ignition system :- It is an critical component of an internal combustion engines that provides the spark needed to ignite the fuel-air mixture in the engine cylinder. Without an ignition system the engine would not be able to start or run.

- The basic component of an ignition system includes a spark plug, an ignition coil & a control module. The spark plug is responsible for creating the spark that ignites the fuel-air mixture in the engine cylinders. The ignition coil is a high-voltage transformer that converts the low voltage electrical signal from the control module into a high voltage signal needed to create the spark. The control module regulates the timing and duration of the spark across that



M. B. M. UNIVERSITY JODHPUR

Date 13/4/23

Experiment No. 6

P. No. 43

Name Zahid Khan

Faculty No. 20/24765

Class II Sem

Batch. A4

it occurs at the correct moment in the engine's combustion engine.

Types of ignition system:-

- (1) Magneto ignition system
- (11) Transmissistorised assisted contact ignition system (TACIS)
- (14) High voltage capacitive discharge ignition system (HVCDIS)

Construction & working:-

(1) Magneto Ignition system:-

A magneto ignition system is a type of Ignition system used in some internal combustion engines. Particularly small engines such as those found in motor cycles, small boats & lawnmowers.

The basic components of a magneto ignition system include a permanent magnet, an armature, and a set of breaker points. The permanent magnet is mounted on the engine flywheel & rotates as the engine runs. The armature is stationary coil of wire that surrounds the magnet, & the breaker points are located between the magnet & the armature.

As the magnet rotates it creates a magnetic field that induces a current in the armature coil. This generated current is then sent to the breaker points, which open



M. B. M. UNIVERSITY JODHPUR

Date 13/4/23

Experiment No. 6

P. No. 44

Name Zahid Khan

Faculty No. 20/24765

Class BE III Sem Batch. A

4 close as the magnet rotates. The opening & closing of Breaker points create a high voltage electrical signal that is sent to the spark plug to ignite the fuel air mixture in the engine's cylinders.

Advantages, 1) their simplicity & reliability
2) not Required external power source
3) not affected Battery failure

- Major drawbacks of conventional (Battery / magneto) Ignition system :-

- (1) Excessive of CB points. ~~None~~ inductive current interruption effects left or CB points / replacement (if condenser is weak)
- (2) The arm of CB point bounces on high speed. but magneto field not is developed.
- (3) At low speed the dwell period is more current drawn from battery slow opening of C.B point hence low secondary voltage (~~dist~~ is small).
- (4) If 300-400 sparks/sec then it is ~~not~~ ^{unsatisfactory} if a bike will be operate about 900-12000 rpm. going less run above this speed, coil ignition system is not satisfaction for them.



M. B. M. UNIVERSITY JODHPUR

Date 13/4/23

Experiment No. 6

P. No. 45

Name Zahid Khan

Faculty No. 20/24765

Class II Sem

Batch. AY

Electronic - Ignition :-

- (1) Uses solid state device
- (2) Transistor & thyristor are ~~suggested~~ ^{suggested} & compact devices
- (3) Can operate as switch at much higher speeds required for ignition system
- (4) Transistor switching is much faster (0.45 mm to 0.65 mm)

Important :-

- (1) Spark plug gap (0.5 to 1mm) [for $\times$ gap 0.1mm - 300V is req.]
- (2) Two plugs can be in one cylinder. Both may fire together or one slightly earlier than other
- (3) It reduces emissions & slightly power increases.
- (4) Sparking Voltage $\propto$ Compression ratio - high ratio means high voltage is required.
 $\Rightarrow \times$ gap of plugs $\rightarrow$ high gap means high voltage is required.

For example :-

No of cycle

Firing order

2

1, 2

3

1 3 2 (maruti)

4

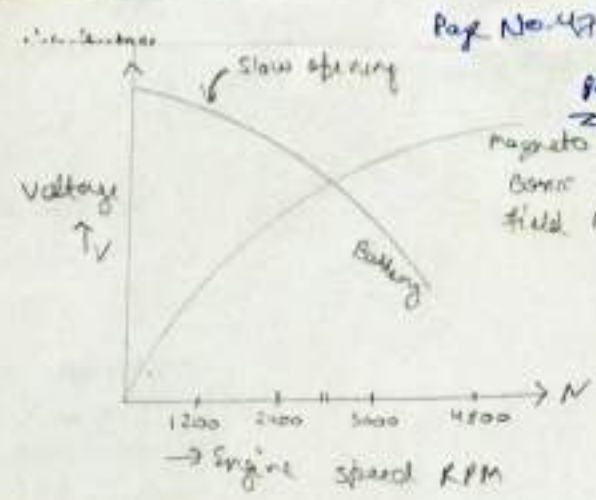
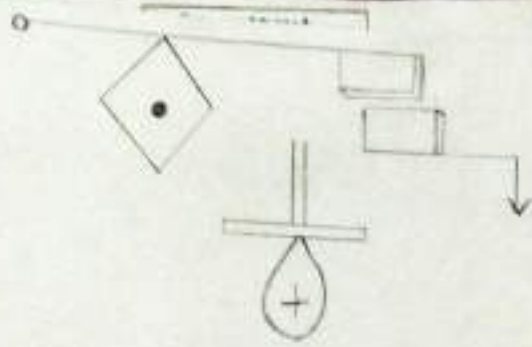
1 4 3 2, 1 3 4 2

6

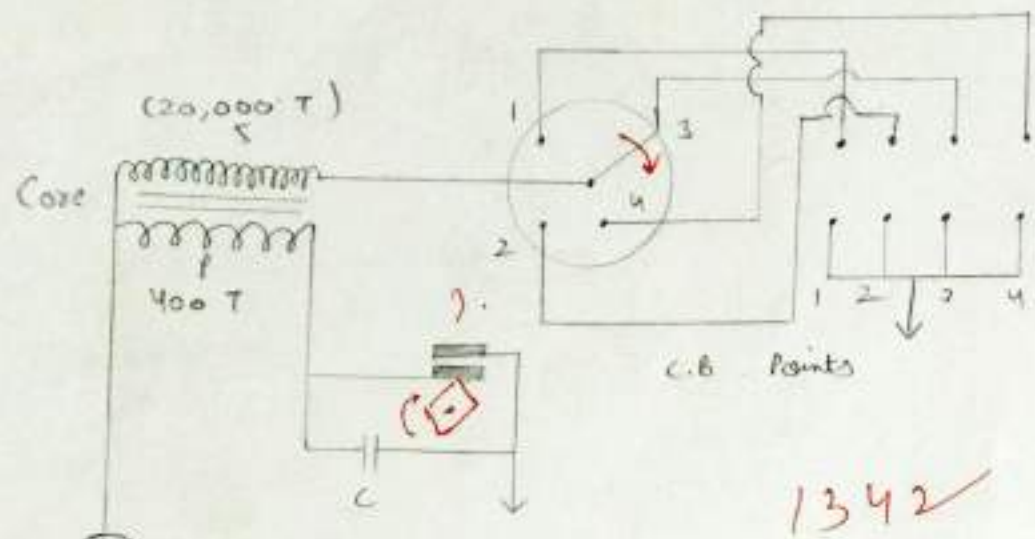
1 5 3 6 2 4, 1 4 2 6 3 5

8

1 6 2 5 8 3 7 4, 1 2 7 3 6 5 4 2

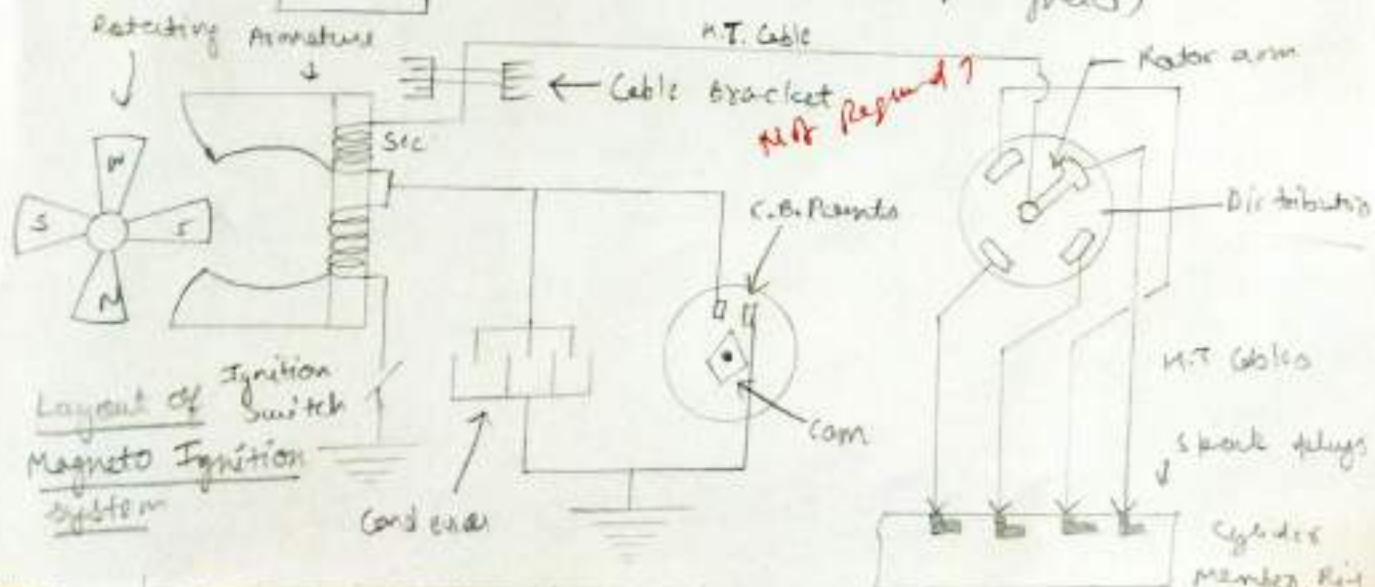


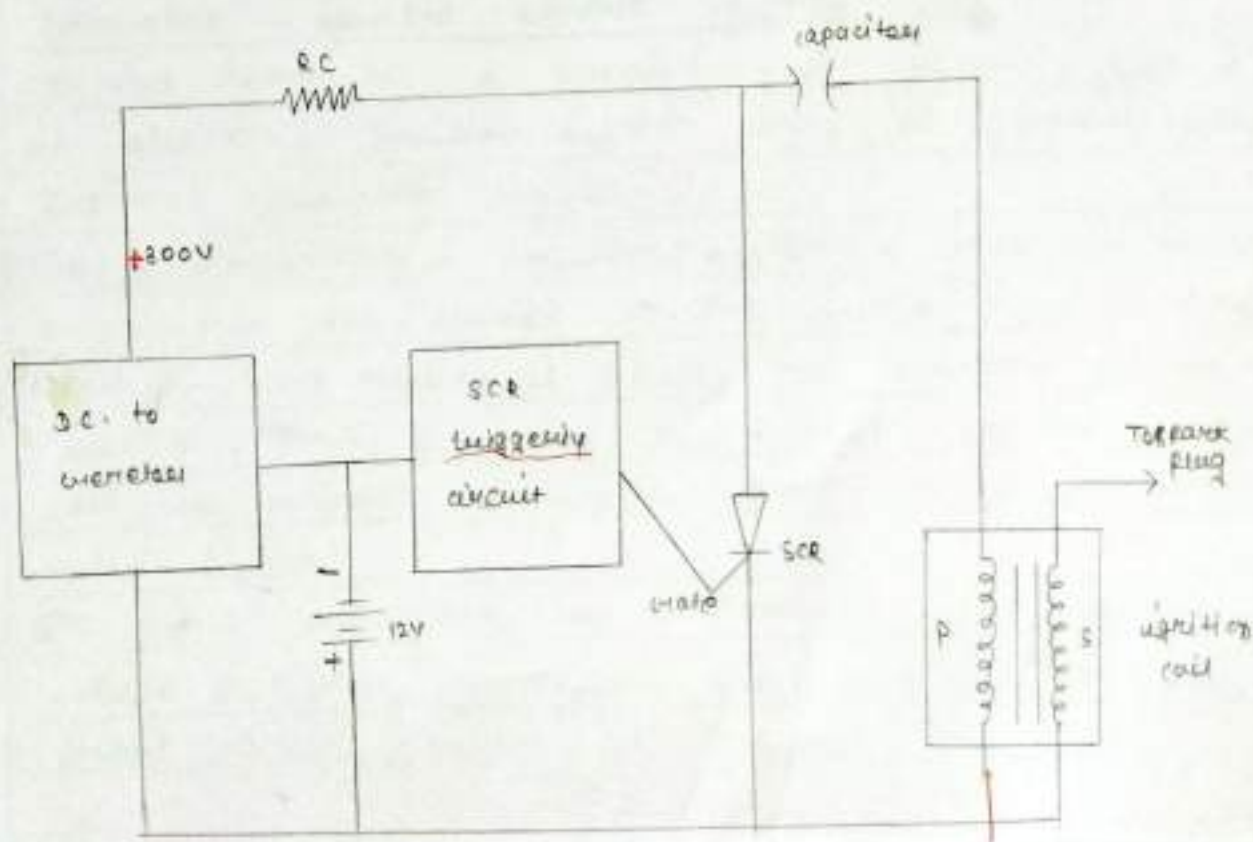
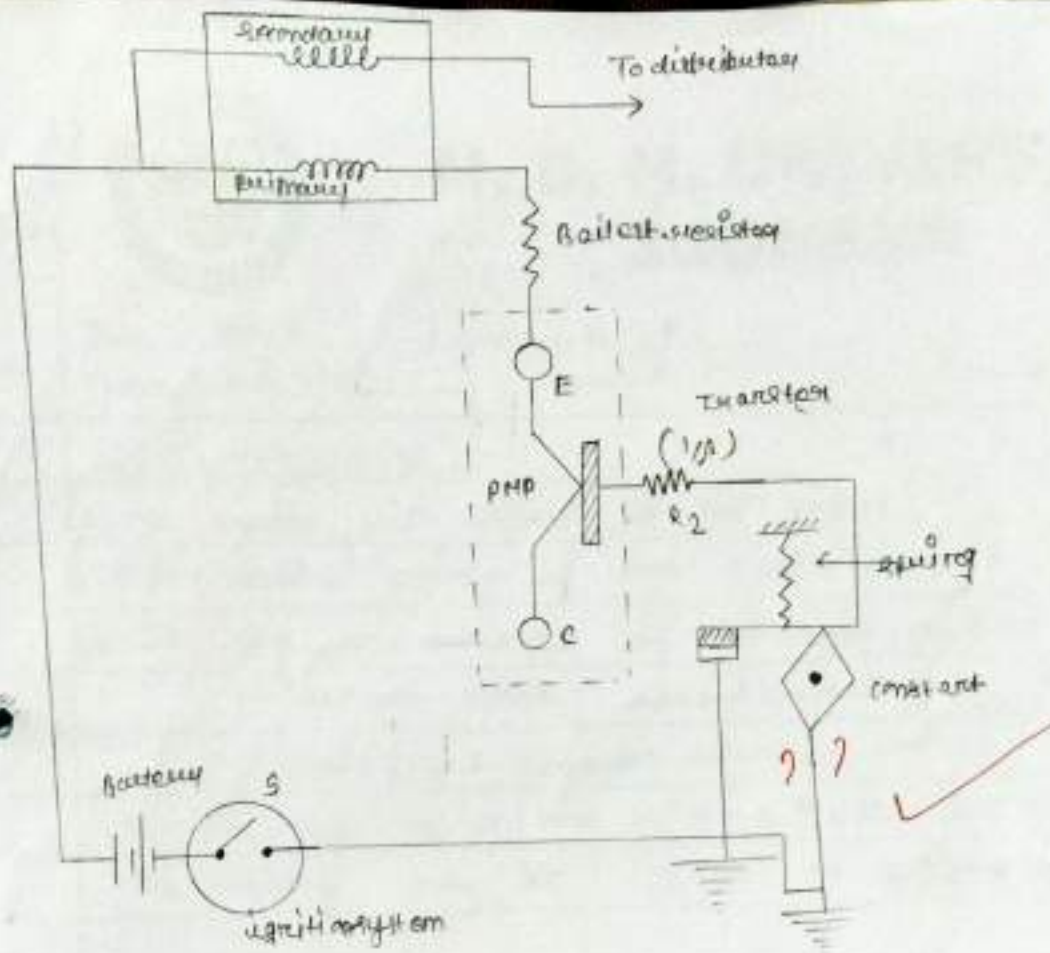
Page No. 48
Zahid Khan
Magnetos
Cover of Full
field nuts develop



1342 ✓

Ignitions - system (Coil & magnetos)







M. B. M. UNIVERSITY JODHPUR

Date 13/4/23

Experiment No. 6

P. No. 48

Name Zahid Khan

Faculty No. 20/24763

Class BE 3rd Sem Batch 174

spark advantage :-

- (1) It varies with speed of the engine.
- (2) For scooter engines it has fixed advance
- (3) It can be achieved by following methods.

- Vacuum spark advance method

- Centrifugal spark advance method

for a gap of 0.1 mm - 300 volts is needed

The voltage may be 10,000 to 20,000 volts.

② Transistor - assisted contact ignition system:-

It also known as a transistorized ignition system is an electronic ignition system used in gasoline-powered Internal combustion engines.

In a transistorized ignition system a transistor is used to switch the current to the ignition coil on & off instead of using mechanical points. the transistor acts as a switch opening & closing the circuit to the ignition coil in response to signals from the engine control module (ECM).

The basic components of a Transistorized ignition system include (1) Battery (2) Ignition coil (3) Transistor (4) Engine control module (ECM) (5) Sensors.



M. B. M. UNIVERSITY JODHPUR

Date 13/4/23

Name Zahid Khan

Experiment No. 6

Faculty No. 20/24765

P. No. 49

Class BE III Sem Batch 1st

When the engine is running the ECM sends signals to the transistors to turn the current to the ignition coil on & off at the correct time to create a spark in the engine cylinders. This spark ignites the fuel-air mixture causing the engine to run.

Advantages :-

- (1) More reliable operation
- (2) Improved performance
- (3) Lower maintenance.

(3) High voltage Capacitive discharge Ignition system:-

It is also called a capacitor discharge ignition system (CDI) is an electronic ignition system used in gasoline powered internal combustion engines it generates high-voltage electrical sparks to ignite the air-fuel mixture in the engine cylinders.

The basic components of a CDI system includes

- (1) Battery (2) charging circuit (3) Trigger circuit
(4) power switch (5) Ignition coil

When the engine is running the charging circuit charges the capacitor to a high-voltage when the



M. B. M. UNIVERSITY JODHPUR

Date 13/4/23

Experiment No. 6

P. No. 50

Lodha

Name Zahid Khan

Faculty No. 20/24765

Class BE III Sem Batch AM

trigger circuit detects that the spark plug is ready to fire, it sends a signal to the power switch, which discharges the capacitor into the ignition coil to generate a high voltage spark.

- * Discussion:- Overall, advanced ignition system for SI engine are focused on improving the efficiency & emission of gasoline - powered IC engine. These system use innovative technologies, & Compression ignition to improve the combustion process & reduce emission

21/4/23



M. B. M. UNIVERSITY JODHPUR

Date 20/4/23

Experiment No. 7

P. No. 51

Name Zahid Ikhon

Faculty No. 20/24705

Class III Sem. Batch A4

* Object:- To study advanced fuel ignition system.

* Theory:-

- Fuel injection is a method of delivering fuel to an internal combustion engine. In a fuel injection system, fuel is sprayed into engine cylinders.
- Fuel injection system have largely replaced carburetors in modern engines because they offer more precise control over the amount of fuel delivered to the engine.
- There are some causes to replace carburetors are:
 1. Carburetor fails to maintain A/F ratio at all conditions.
 2. There is resistance to flow of mixture by throttle valve & venturi also obstructs.
 3. As of these we use superchargers (engine shaft) & turbochargers (turbine - Exhaust gas).

* Merits of injection system:-

1. Higher volumetric η as no venturi restriction. (Air intake $\uparrow$).
2. Spot for preheating of air fuel mixture is avoided.



M. B. M. UNIVERSITY JODHPUR

Date: 20/4/23

Experiment No. 7

P. No. 52

Name: Zahid Khan

Faculty No. 20/24765

Class: BE VI Sem Batch: A4

3. Acceleration response is better.
4. Atomization is greatly improved.
5. Pollution is reduced (by ECU).
6. Precise fuel distribution at all running conditions with multipoint injection.

→ Demerits of injection system:-

1. Initial cost high, maintenance cost is high.
2. Power electrical or mechanical is required.
3. Power bulky & heavy.

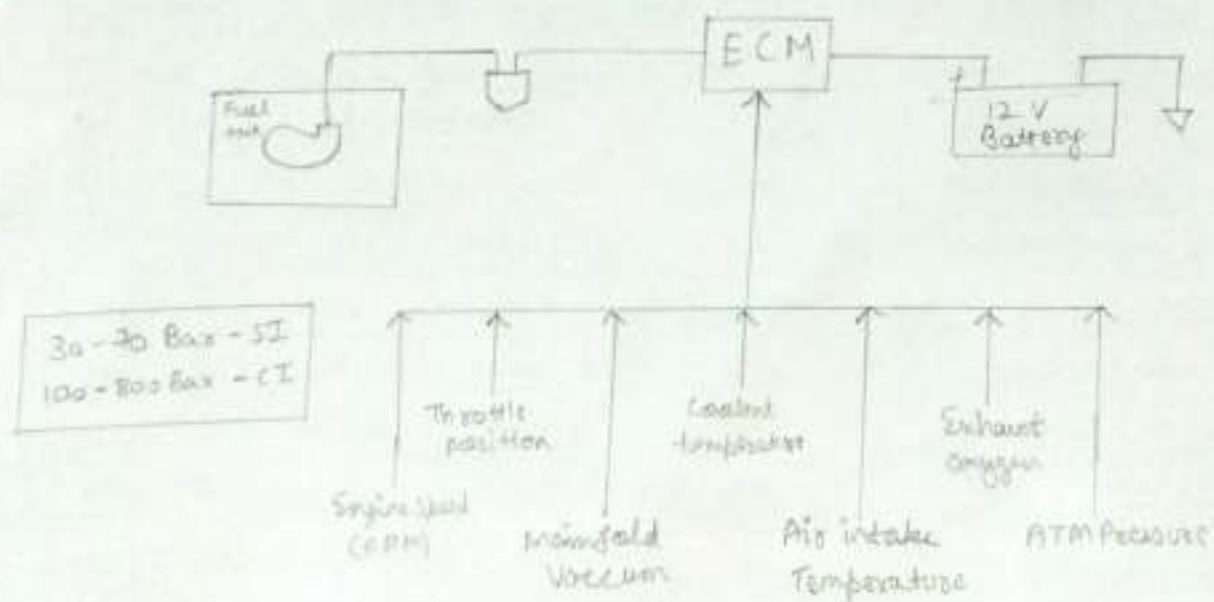
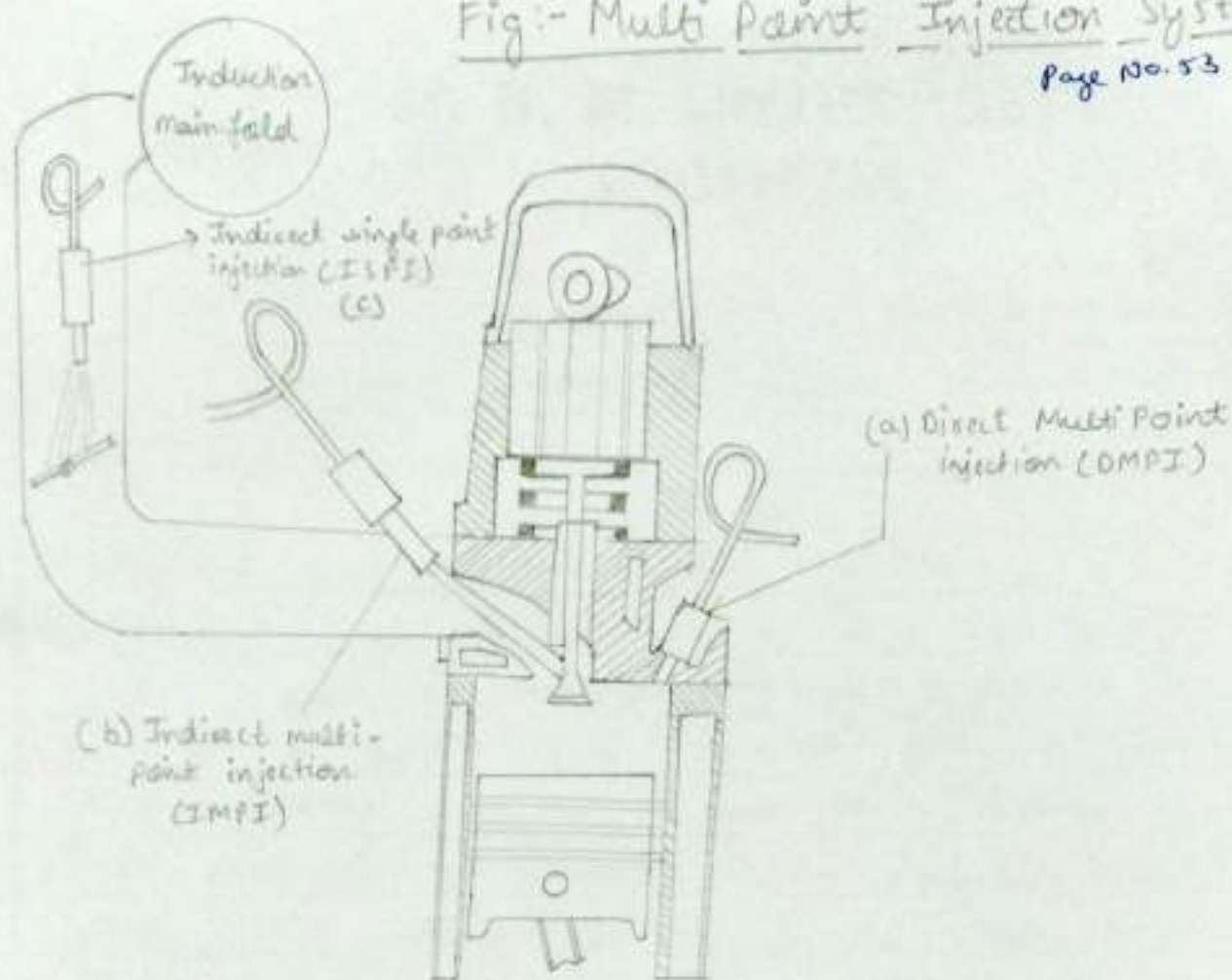
→ Electronic Control module (ECM):-

An ECM is an electronic device that controls various functions of an internal combustion engine. The ECM receives input from sensors throughout the engine, such as oxygen sensors, mass airflow sensors, and throttle position sensors, & uses this information to calculate the optimal fuel delivery, ignition timing & other engine parameters.

The ECM uses microprocessors to process the incoming sensor data & control the various engine functions. It communicates with other vehicle systems such as transmissions, anti-lock braking system, & emissions control

Fig:- Multi Point Injection System

Page No. 53





M. B. M. UNIVERSITY JODHPUR

Date 20/4/23

Experiment No. 7

P. No. 54

Lodha

Name Zahid Khan

Faculty No. 20/24765

Class BE III Sem. Batch. 14

system to ensure smooth & efficient operation of the vehicle.

→ Multipoint fuel injection system (MPFI):-

- Multi-point fuel injection system is a type of fuel injection used in modern internal combustion engines. In a MPFI system, fuel is injected into each section cylinders intake port as opposed to a single injection point used in a throttle body injection (TBI) system. This allows for more precise fuel metering & better engine performance.
- The MPFI system consists of several components, including a fuel pump, fuel filter, fuel pressure regulator, fuel injectors, and an engine control unit (ECU). The fuel pump draws fuel from the fuel tank & delivers it to the fuel injectors at a specified pressure.
- The fuel injectors are located in the intake manifold & spray fuel directly into the intake port of each cylinder. The amount of fuel injected is controlled by the ECU, which takes into account various sensors input such as engine speed, throttle position & air temp.
- The ECU also monitors the oxygen sensors to ensure that the air/fuel mixture is optimal for engine per-



M. B. M. UNIVERSITY JODHPUR

Date 20/4/23

Experiment No. 7

P. No. 55

Name Zahid Khan

Faculty No. 20/24765

Class BE II Sem Batch 94

- Performance and emission control.

→ Common Rail direct injection system (CRDI):-

A common rail direct injection system is a modern fuel injection system used in diesel engines. The system operates at high pressure upto, 2000 bar & delivers fuel to each engine cylinders through a common fuel rail.

The CRDI system consists of several key components, in ordinary a high pressure fuel pump, a common fuel rail, fuel injectors, pressure sensors, & an electronic control unit.

The high-Pressure fuel pump draws fuel from the fuel tank & delivers it to the common fuel rail at high pressure. The common fuel rail distributes the fuel to each fuel injector, which then spray the fuel directly into the combustion chamber of each engine cylinder.

The ECU control the fuel pressure & injection timing, based on various sensors input such as engine speed, load, and temp. The pressure sensors in the fuel rail & injectors provide feedback to the ECU, allowing it to adjust the fuel pressure & injection timing for optimal

Figure:- Individual pump injection system

Page No. 56

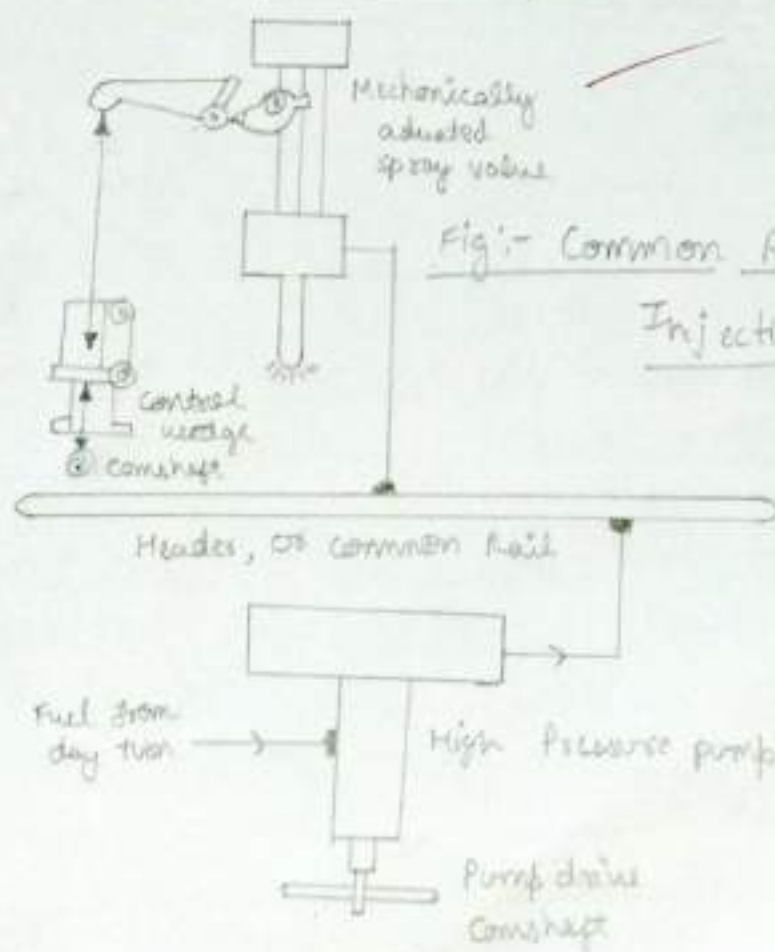
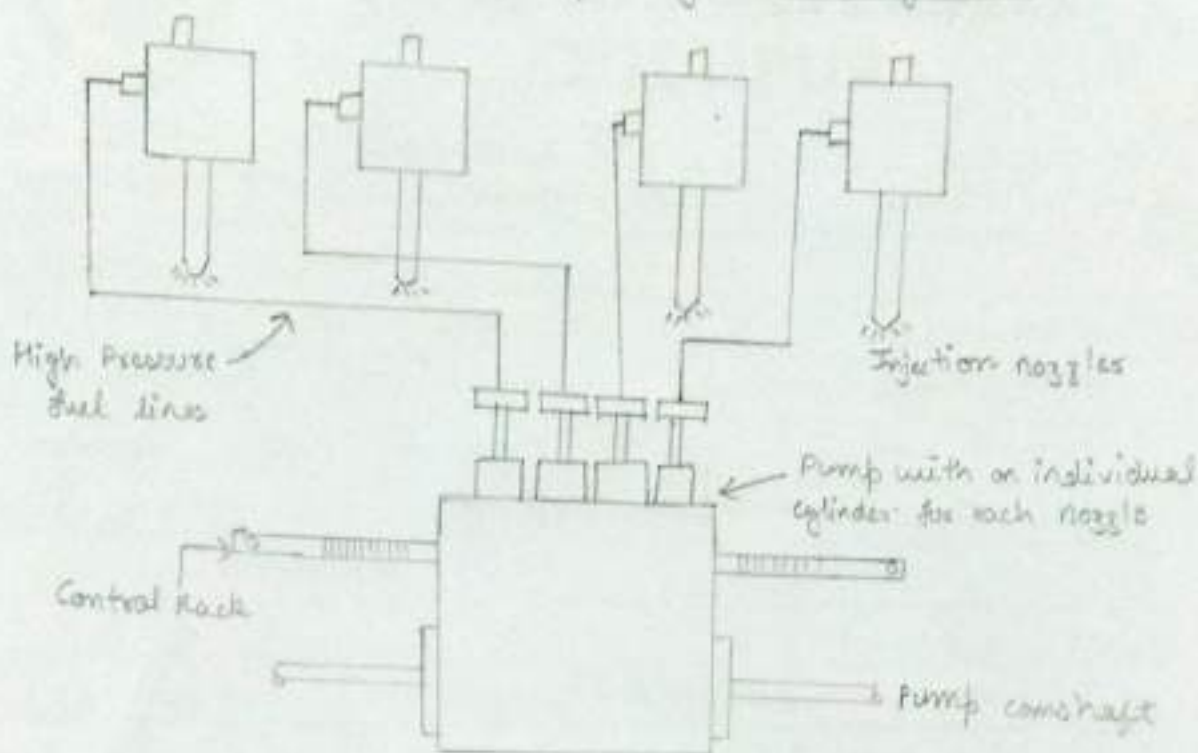


Fig:- Common Rail Direct Injection System



M. B. M. UNIVERSITY JODHPUR

Date 20/4/23

Experiment No. 7

P. No. 57

Name Zahid Khan

Faculty No. 20/24765

Class DE II Sem Batch AY

engine performance.

The CROI system provides several advantages over traditional diesel fuel injection systems. The high pressure fuel delivery & precise injection timing result in better fuel efficiency, increased power output & reduced emissions.

→ Cut section CROI engine (4 cylinder 4 stroke) :-

- | | |
|----------------------|--------------------------|
| (1) Water Pump | (5) oil pump |
| (2) Engine Block | (6) Connecting rod |
| (3) Cam | (7) high pressure fuel |
| (4) Common Rail | (8) chamber |
| (9) Engine Plate | (18) Exhaust |
| (10) Head | (19) Insulator pipe |
| (11) Timing belt | (20) Vacuum sensor |
| (12) filter oil | (21) oil pressure switch |
| (13) Flywheel | (22) Engine heater |
| (14) Injector nozzle | (23) Timing sensor |
| (15) Excess diesel | (24) Piston |
| (16) Turbo charger | |
| (17) suction air | |



M. B. M. UNIVERSITY JODHPUR

Date 4/5/23

Experiment No. 8

P. No. 58

Name Zahid Khan

Faculty No. 20/24765

Class B.E. II sem Batch AY

* Object :- To study various automotive emissions, their norms & control system (Exhaust gas recirculation, catalytic converter).

* Theory :-

- Pollution :- Pollution is defined as introduction of harmful material into environment these harmful materials are called pollutants & they damage the quality of water, air & land.

- AQI - Air Quality Index - O_3 , PM, CO, SO_2 , NO_x , Hydrocarbon
(Measure of Air pollution) ↓
Pollutants

strict test emission norms were initiated in the world first in California USA.

- a) Major Exhaust emission from CI engines is (Particulate Matter)
- b) CO is most dangerous because it is poisonous.
- c) Initially lead was added to petrol to reduce knocking.
- d) EGR → Exhaust Gas Recirculation reduces NO_x , But they reduce thermal η & increased hydrocarbon (HC).



M. B. M. UNIVERSITY JODHPUR

Date 4/5/23Experiment No. 8P. No. 59Name Zahid KhanFaculty No. 20/24765Class BE III SemBatch Am

* The two devices are used to reduce pollutions are:-

EGR (Exhaust gas Recirculation):-

- Used for Diesel & Petrol Engines.

- Excessive NO_x forms when peak temp^s exceeds 1927°C .

To lower the combustion temp^s many engine use EGR system it recirculates a small metered amount (6 to 13%) of the inert exhaust gas back into the intake manifold. The cooler exhaust gas absorbs heat from the much hotter combustion process. This reduces peak combustion temp^s & hence leaves the formation of NO_x . (NO_x is $\propto$ to peak comb. temp^s).

Controls of Emission from S.I. Engines (Main are HC, CO, NO_x):-

These are two methods:-

- (1) To reduce the formation of pollutants by redesigning the engine system, fuel system, cooling & ignition system.
- (2) By destroying the pollutants after these have been formed.

(A) After burner of Exhaust gas:-

(B) Catalytic converter

+ pressurized cooling:-

- ① Usually a P of 2 bar (upto) is used, the boiling pt of water 1.5 bar = 110.4°C ③
2 bar = 120.2°C

- It should have high specific heat (mcvdt)

→ In modern practice engine is run at a temp. as high as possible, generally at 2 bar. to increase thermal η , in pressure cap a safety relief valve & a vacuum valve is used for safety purpose.

* pump ko pura open kyu kholkar rakhte hai
railway workers ko (Ethylene glycol) (infinite type)
↓ (woxy interpart) p-xh

min consumption jab pump band ^{has deta} hai
(at fully closed)

→ Anti rusting (Knock) (matras me)

① Objective

② Apparatus zar ^{→ protection}

③ Theory → coolant, use, cooling water distribution

④ Description →

① water Air type coolant

② water type

③ cooling type

⑤ Discussion & Best (compare)



6/4/23
3

Objective:- To study various ^{eng}lubrication methods
(- feed mech system, mechanical lubricator, splash system / pressure - also study lubrication of piston rings)

Zehid Khan 8

- ① To study engine starting motors (DC motor) & driving arrangement (Bendix drive).

Expt-3. Part ①

Theory

Theory:- Whenever 2 surfaces have relative motion the friction will cause wearing of parts. the loss of power generates heat depending on load & coeffⁿ of friction (μ).

Long life depends on proper lubrication. Sometimes lubricant is called the life blood of machine. mineral oils are commonly used for lubrication. (lubricating)

objects:- ① To cool & remove heat from moving parts

② Seal:- To seal b/w piston & cylinder walls.

③ Clean:- To clean & carry out abrasive dust & metallic particles $\rightarrow$ protection

④ Noise Reduction:-

⑤ Corrosion:- It prevents corrosion. disposal $\rightarrow$ only air

properties:- Cheap, low consumption rate & good cold starts prop.

types:- solid, semisolid, liquid $\rightarrow$ mineral oils

graphite
milk stone
brush choke
molybdenum
sulphide

various types of grease
(these are used for high & slow speed)

Chips T & P

⑥ Methods of lubrication (5)

① Drop feed lubricator ✓

② Ring type lubricator ✗

③ splash system

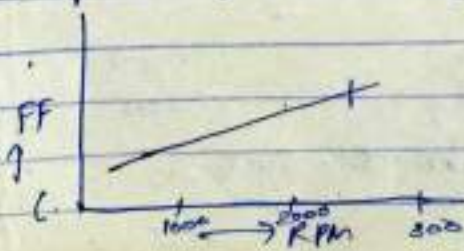
Description

(diesel) lubricants

④ Forced or pressure

⑤ by mixing with fuel.

(Most lubrication) (2 stroke - petrol engine, steam engine)



* Object:- To study advanced fuel injection system (Electronic control multi-point fuel injection system (MPFI), Common rail diesel injection system (CRDI). - diesel

20/4/23

17

Zahid Khan

Q Why we have rejected carburetor?

1. Carburetor fails to maintain air-fuel ratio at all conditions.
2. There is resistance to flow of mixture by throttle valve.
3. Venturi also obstructs
4. There are air surges & turbochargers (turbine-exhaust gas) (engine noise)

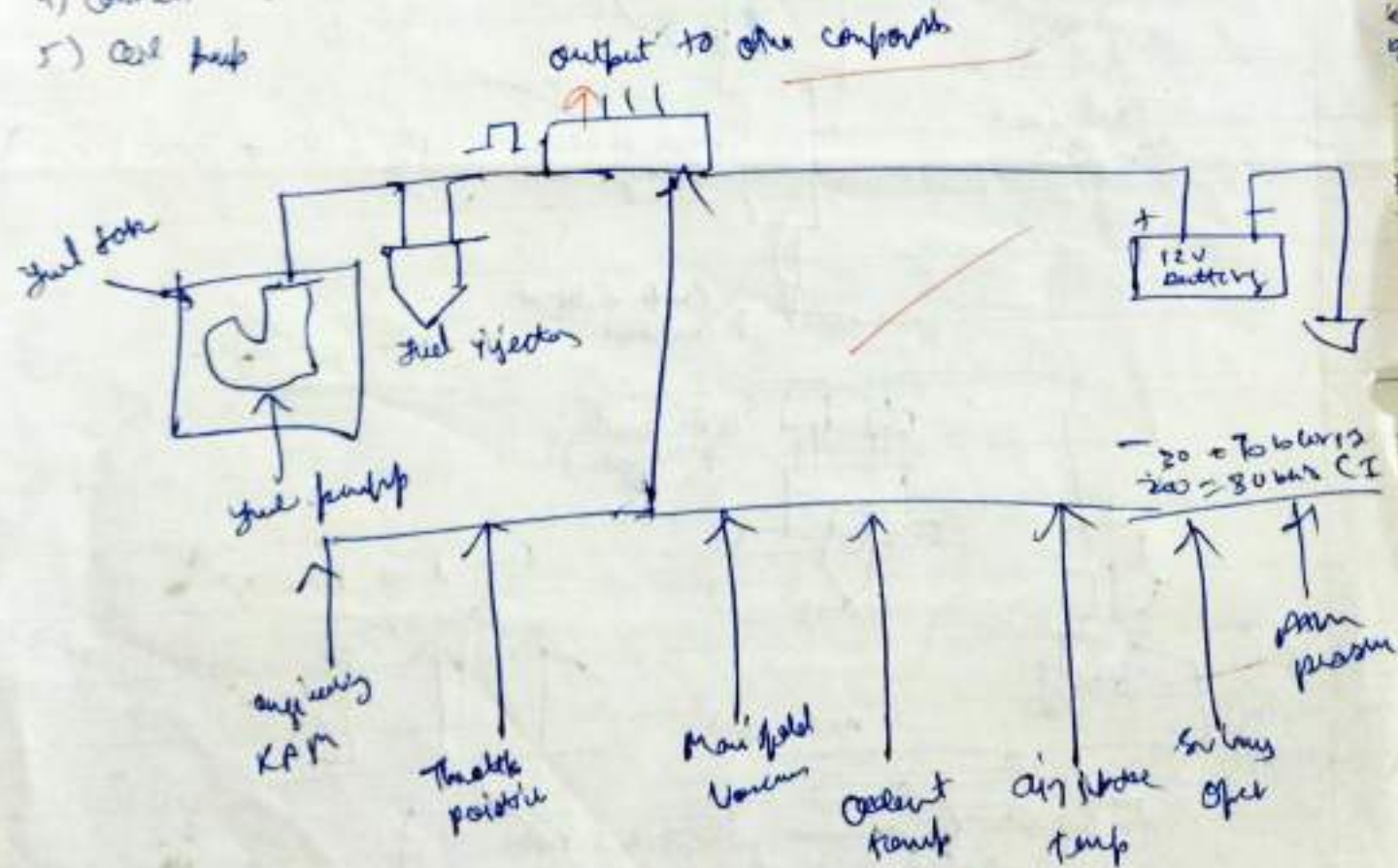
Merits of injection system

- a) High Volumetric η as no venturi restriction. (Air intake T)
- b) Not for preheating of air-fuel mixture is avoided.
- c) Acceleration response is better
- d) Atomization is greatly improved

Cut section of CRDI engine (4 cylinder 4 stroke)

- 1) Water pump
- 2) Engine Block
- 3) Crank
- 4) Common Rail
- 5) Oil pump

Petrol eng's





M. B. M. UNIVERSITY JODHPUR

(12)

Date..... Experiment No..... P. No.....
Name..... Faculty No..... Class..... Batch.....

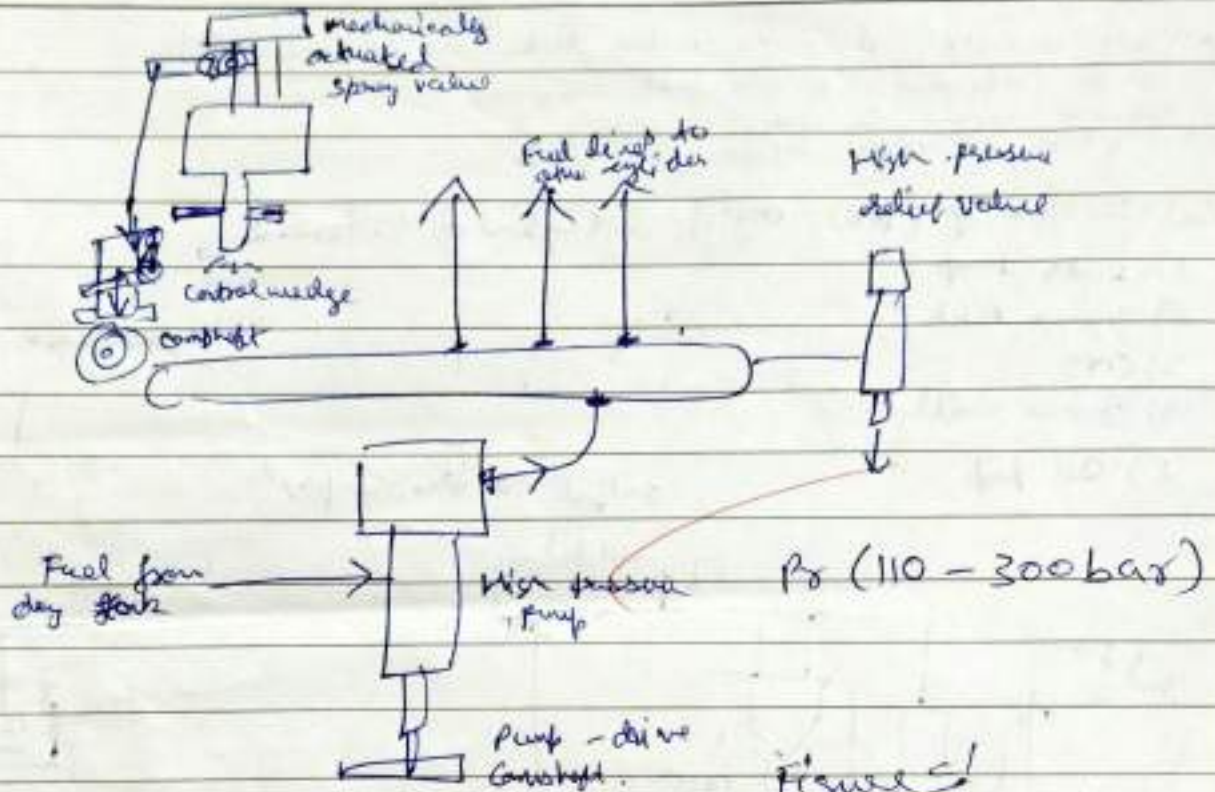


Figure 1

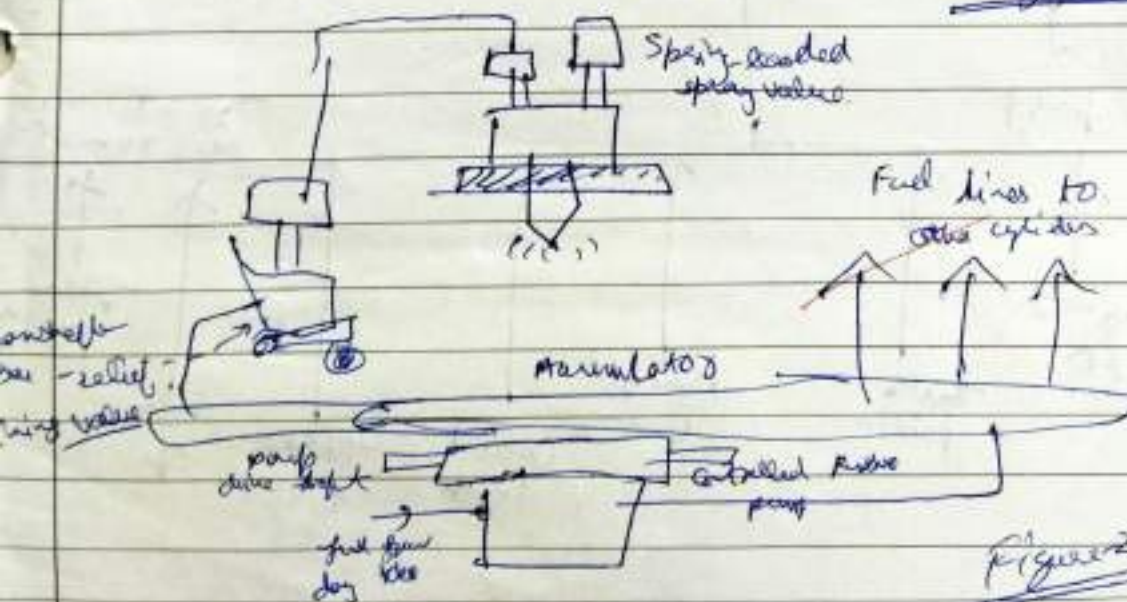


Figure 2

20/4/23

4/5/23
31

Exp-10 → Morse test

$$IP = BP + FP$$

Exp-10

19

$$\text{mech } \eta = \frac{BP}{IP}$$

$$IP = 10, BP = 8, FP = 2$$

$$= \frac{8}{10} \Rightarrow 80\% \text{ or } 0.8$$

Wk 2000

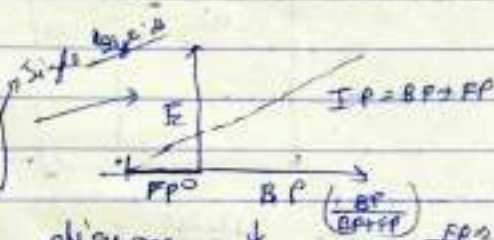
$$BP = 2 \times NT$$

(IP) → (i) William's line

(ii) Motoring test

(iii) By indicated diagram

(iv) Morse test



$$BP = 2, \eta = 50\%$$

$$BP = 8, \eta = 80\%$$

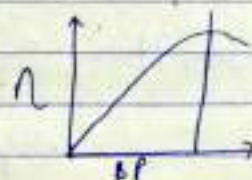
Q If mech is 80%, FP = 2 find

$$\eta = \frac{BP}{BP + FP} = 0.8$$

$$B.P = (0.8)(B.P + 2)$$

$$12 B.P = 16.8$$

$$B.P = 1.4$$



Q If BP = 10, FP = 5, find I.P = 15 (BP + FP)

Q Mech $\eta = 90\%$, BP = 18, F.

$$\eta = \frac{BP}{BP + FP}$$

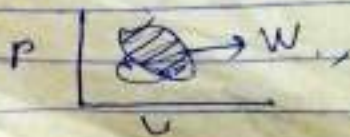
$$FP = 2$$

$$16.2 + \eta FP = 18$$

thick more current
towards lower power
diesel engine
less compression ratio

→ motor's charac waki BP nikala, petrol engine

(iv) indicated dly



$$IP = \frac{P_m \times L \times A \times N}{K}$$

4/5/23 Exp-10 → Morse test

$$IP = BP + FP$$

$$\text{mech } \eta = \frac{BP}{IP}$$

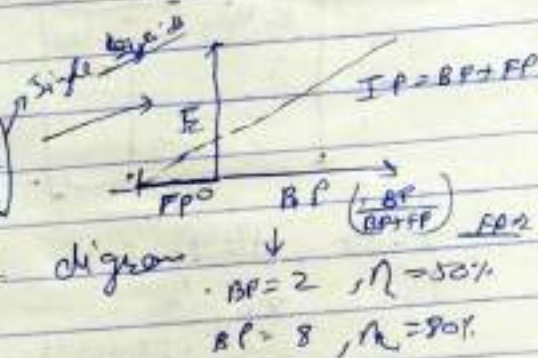
$$= \frac{8}{10} \Rightarrow 80\% \text{ or } 0.8$$

$$IP=10, BP=8, FP=2$$

WN 2000 → degree

$$BP = \frac{2 \times NT}{K}$$

- (IP) → (i) William's line
(ii) Motoring test
(iii) By indicated diagram
(iv) Morse test.



Q If η is 80%, $FP=2$ find

$$\eta = \frac{BP}{BP + FP} = 0.8$$

$$B.P. = (0.8)(B.P. + 2)$$

$$12 B.P. = 16.8$$

$$B.P. = 1.4$$

Q If $BP=10, FP=5$, find $I.P. \Rightarrow 15 (BP + FP)$

Q Mech $\eta = 90\%$, $BP=18$, find

$$\eta = \frac{BP}{BP + FP} = 0.9$$

$$FP = 2$$

$$16.2 + 0.9 FP = 18$$

thick more current
turbine (more power)
diesel engine
(less compression ratio)

→ motor chalo uske BP nikalo, refined engine

(10) indicated HP



$$I.P. = \frac{P_m \times L \times A \times N}{K}$$

theory test Dr. I.P.

Zahid Khan

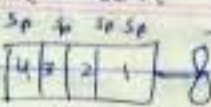
(20)

(10) Mohr's 5-test :-

→ At least 2 cylinders

15

(multi cycle)



(Figure)

1500 rpm = 1500

* Throttle constant

(B)

For note see cont. part 2nd cycle

All cylinder working = 10 kW (dynamic)

→ One by one cylinder is short circuited then run again

(B₁) I cut = 7 kW (B₂) II cut = 7.5 kW

(B₃) III cut = 6.5 kW (B₄) IV cut = 7.0 kW

$$I = 48 - (B_1 + B_2 + B_3 + B_4)$$

IP = 12 > greater than BP = 10

$$\text{Mean } \eta = \frac{10}{12} \times 100 \Rightarrow 83.34$$

→ Like here we take force na. Post

weakest = 3 ✓ strongest = 2

All cylinder works 20 kW (1) 16 kW, (2) 15 (3) 13 (4) 12

16 I.P = 24

$$\eta = \frac{20}{24} \Rightarrow 83.33\%$$

4/5/23

Exp-8

(Pollution)

BSC

Practical Standards 1

It is defined as introduction of harmful material into the environment. These harmful materials are called pollutants. The quality of air, water and land.

→ AQI (Air Quality Index) → measure of air pollution

Pollutants → O_3 , PM (particulate matter), CO, SO_2 , NOx, Hydrocarbons

→ Strict test emission norms were initiated in the world ① is California USA.

→ Major exhaust emission from I.C. engine is PM.

→ CO is most dangerous (because it is poisonous).

→ Initially lead was added to petrol to reduce knocking, EGR

↓
reduces NOx, but reduces thermal η & increases hydrocarbons.

→ Catalytic converter (Figure-1) (IV & VI)
→ EGR (Exhaust gas recirculation) (Figure-2)
pollution table

→ provides η ② HC fuel

Submit → Index → S.No, Exp, Date / Page No, check
Remarks

Exp-9

Exp → Variable Value finding

Zohid Khan
(22)

4/5/23

↑ ↓ ↑ ↓
S C P E



reset → 6 position
(fuel start → fuel stop)

Calculated
PE → 5 point

42 MPa → 6 to 10
avg 7 to 10

(nearly same) DE → 11 to 22
avg 16 to 18

Compression
section

PE
DE → specific gravity → .75
→ .84

BSFC = .3 kg / kWh

BSFL = .27 kg / kWh

mass of fuel per hr
BP

A/F ratio
petrol/engine → 10 to 14 (14.7:1)

A/F DE → 18 to 200

Figure 2

FCX

2 stroke → fuel waste
Diesel cycle → NO waste



M. B. M. UNIVERSITY JODHPUR

Date..... Experiment No..... P. No.....
Name Zahid Khan Faculty No. 20/24765 Class BE II sem Batch Au

Index

S.No.	Experiment	Page No.	Date	check
1	To study different types of Carburetors	1-10	2/3/23	checked
2	To study various cooling methods	11-17	9/3/23	checked
3	To study various engine lubrication system	18-27	6/4/23	checked
4	To study engine starting motors	28-34	6/4/23	checked
5	To study DC generator	35-41	13/4/23	checked
6	To study advanced ignition system	42-50	13/4/23	checked
7	To study advanced fuel injection system	51-57	20/4/23	ok
8	To perform mouse test	58-61	4/5/23	ok
9	To study various automotive emissions.	62-66	4/5/23	ok
10.	To study advanced valve timing	66-71	4/5/23	ok



M. B. M. UNIVERSITY JODHPUR

Date 28/03/23Experiment No. 1P. No. 1Name Zahid KhanFaculty No. 20/24765Class 8E III Sem. Batch PM

* Objective :-

- To study different types of carburetor
(a) SU carburetor (b) Zenith carburetor

* Apparatus required :-

- SU carburetor and zenith carburetor

* Theory :-

- A carburetor is a device that mixes fuel and air in an internal combustion engine. The principle of a simple carburetor is to create a venturi effect, which is achieved by narrowing the diameter of the carburetor throat, creating a constriction in the airflow.
- The constriction causes a decrease in pressure, which draws fuel into the airflow through a small nozzle called a jet. The fuel is stored in a float chamber, and a float valve controls the amount of fuel that enters the carburetor.
- As the fuel enters the carburetor, it mixes with the air, creating a fuel-air mixture that is then



M. B. M. UNIVERSITY JODHPUR

Date 23/03/23

Experiment No. 1

P. No. 2

Name Zahid Khan

Faculty No. 20/24765

Class BE III Sem Batch 1st

delivered to the engine.

- A throttle valve, which is often a butterfly valve that controls the air enters the engine, regulates how much fuel is drawn into the carburetor.

In this experiment we have 2 types of Carburetor are:-

(1) SU Carburetor:-

- The SU carburetor uses the engine vacuum to move the piston that regulates the flow of fuel & a butterfly making it a constant depression Carburetor type.
- It features a piston that regulates the flow of fuel & a butterfly valve that regulates the amount of air entering the carburetor.
- The suction produced by the intake stroke elevates the piston while the engine runs, allowing more fuel to enter the carburetor.
- It is noted for its smooth operation & excellent fuel economy.

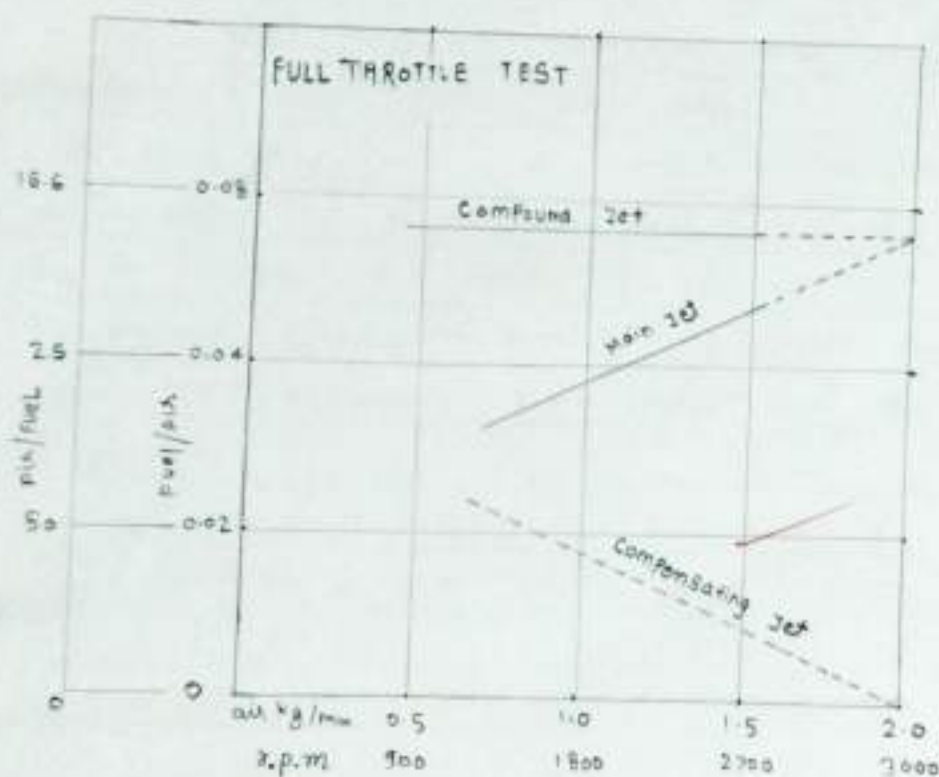


Fig: Full throttle characteristic curve for main, compensating, and compound jets (air-fuel ratio vs air mass flow)



M. B. M. UNIVERSITY JODHPUR

Date 28/03/23Experiment No. 1P. No. 4Name Zahid KhanFaculty No. 20/24765Class BE III Sem Batch PM

(2) Zeinth Carburetor :-

- A form of updraft carburetor, the zeinth carburetor allows fuel and air to enter the engine from the bottom & exit from the top.
- Fuel can flow into the zeinth carburetor through a number of metering orifices that are regulated by a number of valves & springs.
- The manual choke & throttle valve on the zeinth carburetor regulates how much fuel & air enter the engine.
- The zeinth carburetor is renowned for being simple & dependable, and older cars frequently used them.

* Description :-

(1) SU Carburetor :- (Skinner Union)

• Construction :-

(1) Jet Sleeve

(2) Tapered metering needle

(3) Choke piston

(4) Throttle valve

(5) Suction disc

(6) Suction chamber

(7) Atmospheric hose

(8) Light spring

1. Jet sleeve
2. Tapered Metering Needle
3. Choke piston
4. Throttle Valve
5. Suction Disc
6. Suction chamber
7. Suction hole

8. Atmospheric hole
9. Light strip
10. Petrol jet orifice
11. Jet Lever For cold starting
to lower the
jet tube
12. Oil dash pot
13. Piston guided rod slight rise
mix for
acc.
14. Piston
15. Oil cap nut

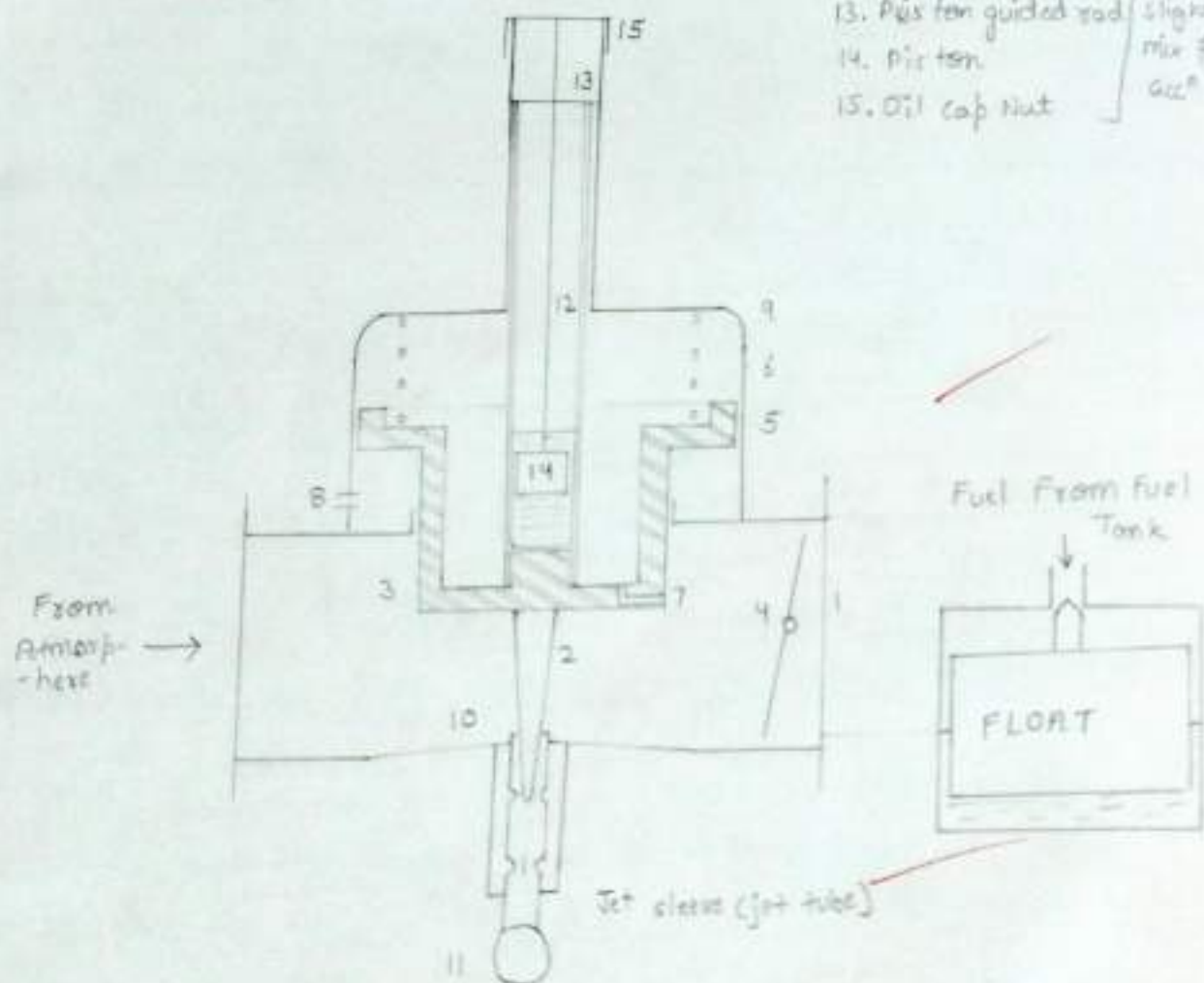


FIG. Schematic diagram of S.U. carburettor



M. B. M. UNIVERSITY JODHPUR

Date: 21/03/23

Experiment No. 1

P. No. 6

Name: Zahid Khan

Faculty No. 20/21765

Class B.E. VI Sem Batch. A4

- | | |
|-------------------------|-----------------------|
| (9) Suction hole | (13) Piston guide rod |
| (10) Petrol jet orifice | (14) Piston |
| (11) Jet lever | (15) Oil cap nut |
| (12) oil dash pot | |

* Working :-

- Air enters the carburetor through the air intake when the engine is running and travel through the venturi, the carburetor thinnest section.
- The main jet & a low pressure area produced by the venturi are used to ~~drawn~~ fuel from the fuel bowl into the air stream. Air & fuel are combined as they pass through the intake manifold & enters the engine.
- In response to the engine Vacuum, the piston in the carburetor body rises & falls. Based on the position of the piston, the ~~needle~~ valve inside the piston assembly control the fuel flow.
- The piston is at the top of the carburetor when the engine is idle or operating at low speeds, which lower the fuel flow.
- The piston descends as the engine speed rises, opening



M. B. M. UNIVERSITY JODHPUR

ENGCOL. Co-op. Stores Ltd., Jodhpur

Date 23/02/23

Experiment No. 1

P. No. 7

Name Zahid Khan

Faculty No. 20/24765

Class BE VI Sem Batch 173

the needle valve & enhancing the fuel flow.

- When there is a sudden acceleration or deceleration, the damper inside the piston assembly helps to stabilize the piston movement & avoid fuel starvation.

- The throttle valve controls the amount of air entering the engine & is adjusted by the driver using the accelerator pedal.

* Use:- Various British Cars, Hindustan Ambassador Cars

(2) Zeinth Carburetor:-

* Construction :-

(1) Main Jet

(6) Throttle

(2) Comp Jet

(7) Slow ramp supply tube

(3) Float chamber

(8) Air bleeding

(4) Narrow restriction

(9) choke valve

(5) well

* Working :-

- The throttle valve, which regulates the amount of air entering the engine, is located at the top of the carburetor, where air enters the carburetor when the engine is operating.

1. Main Jet
2. Lamp Jet
3. Float chamber
4. Narrow Restriction
5. Wall
6. Taper
7. Slow ramp supply tube
8. Air bleeding
9. Choke Valve

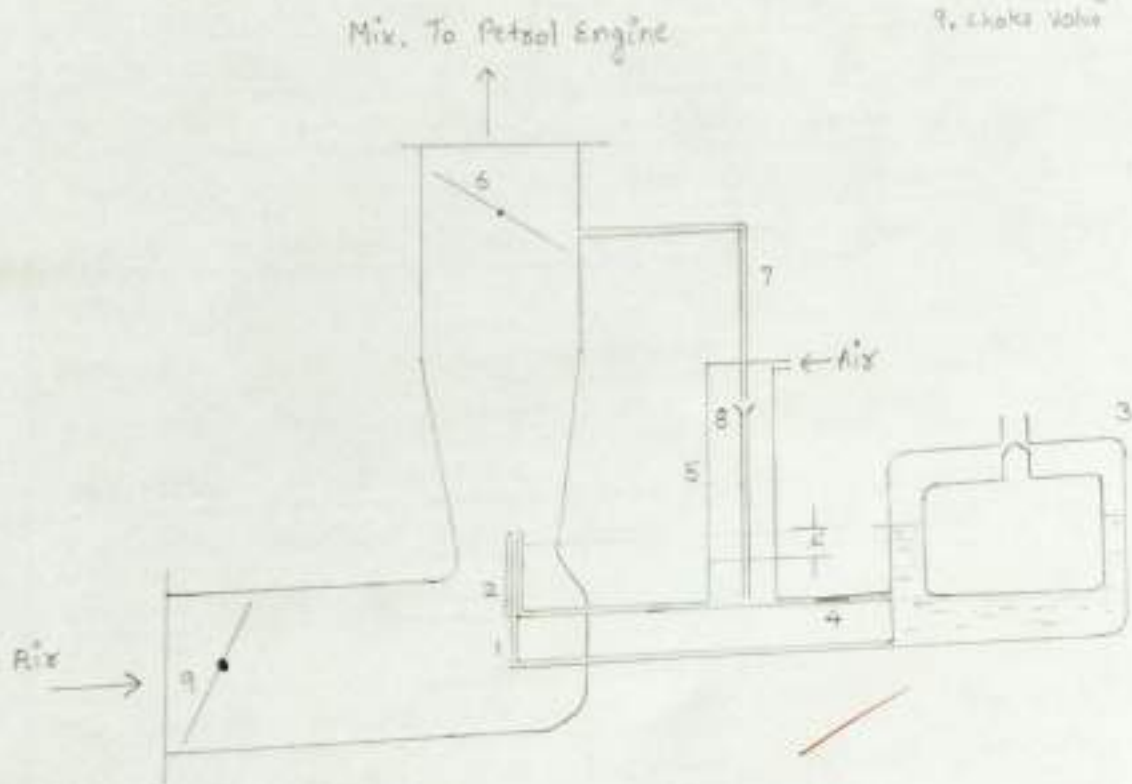


FIG: Schematic diagram of Zenith Carburettor



M. B. M. UNIVERSITY JODHPUR

Date 23/03/23 Experiment No. 1 P. No. 9
Name Zahid Khan Faculty No. 20/24765 Class BE III Sem Batch Ay

- The fuel is drawn from the primary jet & injected into the air stream after the air passes through a venturi, which creates a low-pressure zone.
- The size of the jet and the fuel pressure in the float bowl regulated how much fuel flows through the main jet.
- Moreover, the Zenith carburetor features several metering circuits that control the fuel flow under various operating circumstances.
- The idle circuit, low-speed circuit, and high-speed circuit are some examples of these circuits.

The petrol supply through the main jet continuously increase with the increase of suction due to increase of engine speed.

- Normal Runup:-

Both the jet supply the fuel petrol supply through the M-jet cont. increases with increase in suction and engine speed. At higher speed the supply through comp. jet (the mix becomes leaner).



M. B. M. UNIVERSITY JODHPUR

Date 28/6/23Experiment No. 1P. No. 10Name Zahid KhanFaculty No. 20/24765Class BE VI Sem Batch A4

Startup:-

choke is used. The engine at depression pulls out more petrol ① & ②

Idling:-

Throttle closed supply through ⑦ [main & Comp. jet do not depends, as the suction near the nozzle is not enough.

Uses:- Ambassador, FIAT, Standard & Willy Teks.

Discussion:-

- SU & Zenith Carburetors, in general, are mechanical devices that combines fuel & air to produce a combustible mixture for internal combustion engines. The right air fuel mixture must be delivered to the engine in a variety of operating situations, despite the fact that they function accordingly to different principles.
- SU Carburetor [Constant Vacuum type Variable choke type]
Zenith Carburetor [Constant choke type Variable depression type]

n 9.3.23



M. B. M. UNIVERSITY JODHPUR

Date 9/3/23 Experiment No. 2 P. No. 11
Name Zahid Khan Faculty No. 20/24765 Class BE II Sem Batch A4

* Objective:- To study various cooling methods for IC Engines.
(Air, water & coolant cooling)

* Theory:-

• Coolant and its properties:-

- Coolant is a fluid used to remove heat from a system or a machine, typically in a vehicle or a manufacturing process.
- Its primary purpose is to prevent overheating which can cause damage to the system.

- Some common characteristics of coolant include:-

1. High thermal capacity

2. Low Viscosity

3. High boiling point

4. Corrosion resistance

5. Low freezing point

6. Compatibility

7. Non-toxic

8. Antifreezing / rusting

• Some commonly used antifreeze mixture additives are:-

1. Alcohol

2. Glycerin

3. Kerosene

4. Sugar solution

5. Calcium & magnesium chloride

6. Ethylene glycol

7. propylene glycol



M. B. M. UNIVERSITY JODHPUR

Date 9/3/23

Experiment No. 2

P. No. 12

Lodha

Name Zahid Khan

Faculty No. 20/24765

Class BE III sem Batch FY

• Requirement of cooling:-

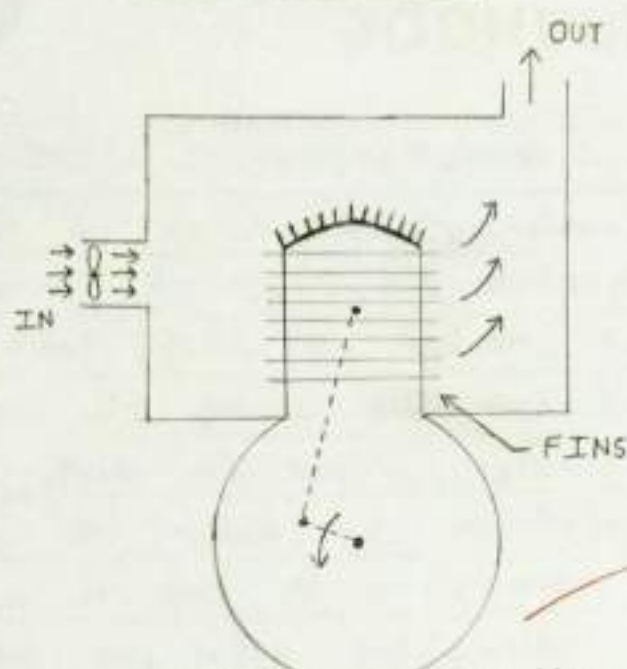
1. Engine Sizer, distortion of Engine block.
2. Hotspot in the cylinders, pre ignition & loss of power.
3. If engine temperature is high then no doubt brake thermal η increases but engine scissors take place.
4. Higher the cooling temp. of water less is the friction because viscosity at high temp. is lower.
5. At high temp. piston will expand considerably and size the liner.
6. Excessive stress in parts sending them useless for further operations.
7. At a higher temperature lubricating oil will evaporate rapidly, piston & cylinder may damage.

* Description:-

1. Air Cooling System:-

- An air cooling system uses air as the cooling medium to remove heat from a machine or system. Common application for this kind of cooling system include tiny engines, electrical devices, and various industrial process.
- A fan is generally employed in air-cooling system to move air across the heated surface of the apparatus.

AIR - COOLING



Small Engine & Aeroplanes

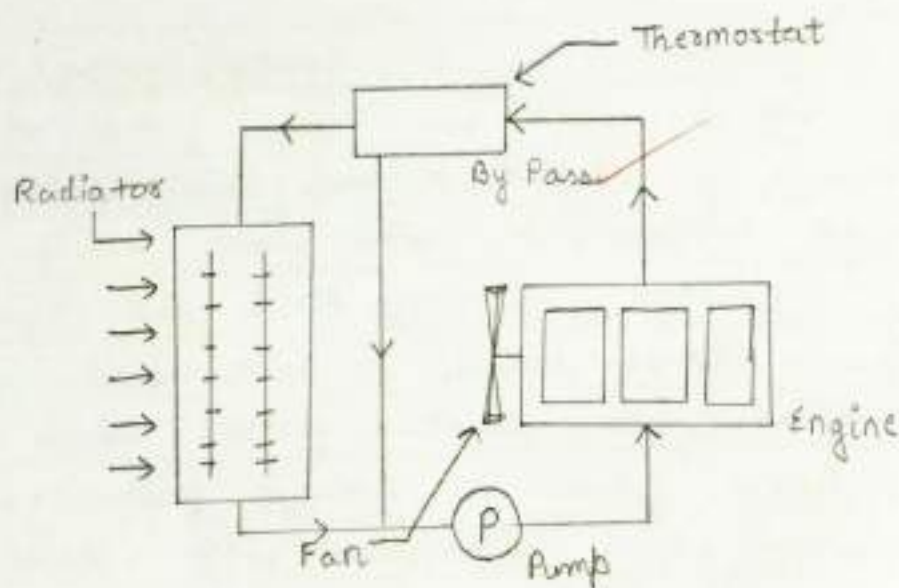


Fig:- Thermostatically controlled Cooling System



M. B. M. UNIVERSITY JODHPUR

Date 9/3/23

Experiment No. 2

P. No. 14

Name Zahid Khan

Faculty No. 20/21765

Class BE II Sem Batch 14

The heated air is subsequently release into the environment making room for cooler air to replace it & further remove heat from the system. The system can stay cool since the process keeps repeating itself.

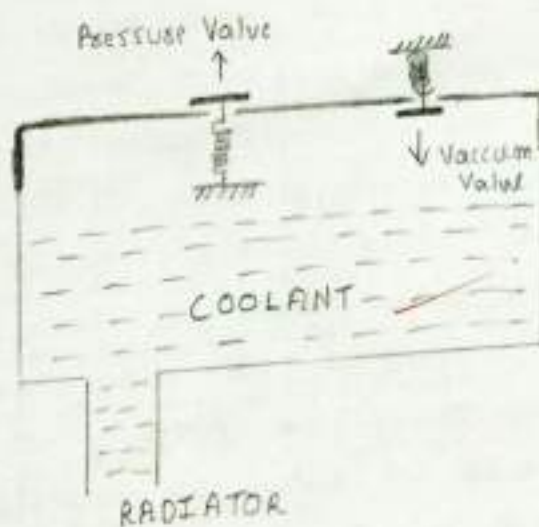
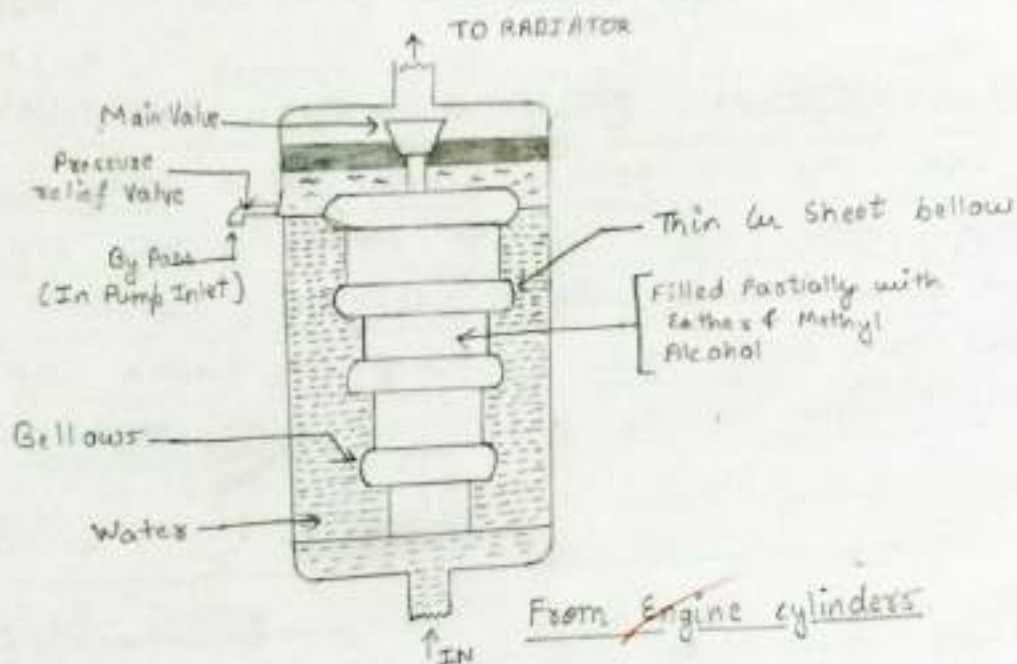
The fact that air-cooling system are easier & less expensive to install and maintain than other cooling techniques is one of their main benefits. Also, they don't need any extra refrigerants or cooling fluids, which may be expensive and possibly dangerous for the environment.

2. Water Cooling system:-

A water-cooling system uses water as the cooling medium to remove heat from a machine or system. Large engines, power plants, data centres & industrial operations frequently employ this kind of cooling system.

A close loop that is linked to the device or system that needs cooling requires water circulations in a water-cooling system. The water circulates through a radiator after absorbing heat from the system, where it is cooled by air or another medium. The device then circulates the cooled water once again to further remove heat.

TYPICAL - THERMOSTAT



Pressurised-cooling-



M. B. M. UNIVERSITY JODHPUR

Date 9/3/23

Experiment No. 2

P. No. 16

Name Zahid Khan

Faculty No. 20/24765

Class BE II Sem Batch 194

Index

- The fact that water-cooling systems are often more effective than air-cooling systems enables bigger cooling capacity & more accurate temperature cooling control. This is one of their key advantages. Moreover, water is a more efficient cooling medium than air because it can ~~absorb~~^{absorb} more heat.
- 3. Pressurized cooling system:-
 - Many contemporary cars and other pieces of machinery employ a sort of cooling system called a pressurized cooling system. The coolant is held under pressurized cooling system, raising its boiling point & enabling it to absorb more heat from the engine.
 - The radiator, thermostat, water pump, and coolant reservoir often make up the pressurized cooling system. The coolant is pumped through the radiator, where it is cooled by air or another medium before being pumped back through the engine block. The purpose of the coolant reservoir is to store any extra coolant & to enable the coolant to expand & contract when the temperature changes.
 - A pressurized cooling system has several benefits, one of which being the increase ability of the coolant to absorb heat from the engine, hence reducing the risk of overheating & engine damage. Also it, aids in preventing the coolant from boiling & evaporating, which might lessen the cooling system's efficacy.



M. B. M. UNIVERSITY JODHPUR

Date 9/3/23 Experiment No. 2 P. No. 17
Name Zahid Khan Faculty No. 20/24705 Class BE III Sem Batch A4

* Discussion:-

- The most ~~eff~~ effective cooling system for a given application relies on a number of variables, included the machinery type, work environment, and cooling requirements. Due to the unique benefits and drawbacks of each cooling system, there is no universally applicable response to this query.
- In general, water cooling systems outperforms air-cooling systems in terms of efficiency & cooling performance, especially when it comes to high speed engines & massive machines. As water is a ~~superior~~ heat-sink than air, engines run more efficiently and last longer when using water-cooling systems, which also offers more accurate temperature control.
- Yet, compared to water-cooling systems, air cooling systems can be easier & less expensive to install & operate, making them more suited for smaller engines and applications where cost is a ~~factor~~. Also, since air-cooling systems do not need a coolant fluid, there is a less chances of leak & spills.

11²
6.4.23



M. B. M. UNIVERSITY JODHPUR

Date 6/4/23

Experiment No. 3

P. No. 13

Name Zahid Khan

Faculty No. 20/24765

Class BE VI Sem Batch AY

* Objective:- To study various engine lubrication methods (feed splash system, pressure feed system, mechanical lubrication, also study lubrication of piston rings)

* Theory:-

(1) Role of lubrication:- Whenever two surfaces have relative motion the friction will cause wearing of parts. The loss of power generates depending on load & coefficient of friction (μ).
- Thus long life depends on proper lubrication, sometimes lubricant is called the life blood of machine.

(2) Uses of lubrication:- Mineral oils are commonly used for lubrication as:-

- (i) Cooling:- To cool & remove heat from rubbing parts.
- (ii) Seal:- To seal between piston & cylinder walls.
- (iii) Clean:- To clean & carry out abrasive dust & metallic parts.
- (iv) Noise reduction:-
- (v) Corrosion:- It prevents corrosion.

(3) Properties:-

- It should be cheap, low consumption rate & good cold starting property.



M. B. M. UNIVERSITY JODHPUR

Date 6/4/23

Experiment No. 3

P. No. 19

Name Zahid Khan

Faculty No. 20/24765

Class BE III Sem Batch AY

* Types of Lubricant :-

(1) Solid:- Consists such as graphite, milk stone, french chalk, molybdenum sulphide.

- Used at high temperature & pressure.

(2) Semi-solid:- Consists various types of grease.

- Generally used for high pressure & slow speed.

(3) Liquid:- Consists various types of mineral oils.

* Figure 1:- As Temperature increases viscosity decreases which reduces friction losses upto a certain temperature.

- As Temperature goes above 90°C the local oil film destroys & metal to metal contact increases which cause more & more friction losses.

* Methods of Lubrication:-

(1) Splash Lubrication system:-

This system is used on small four stroke stationary engines.

In this case the cap on the big end bearings of connecting rods are provided with scoops which, when the connecting rod is in the lowest position, just dip into oil through

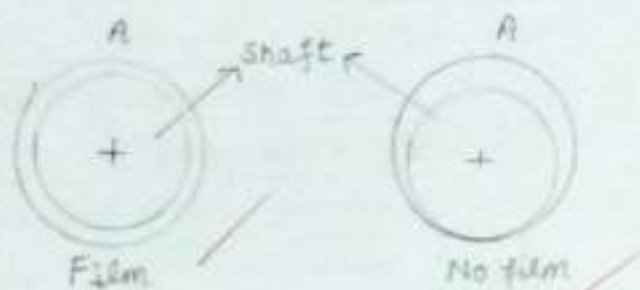
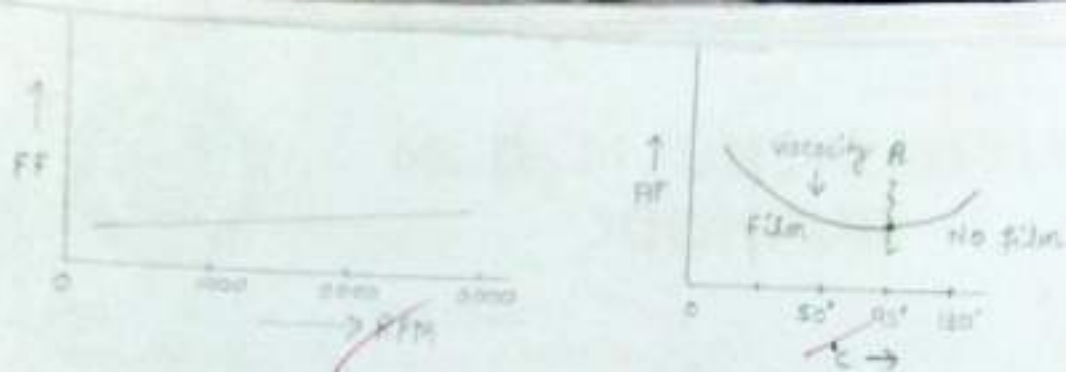
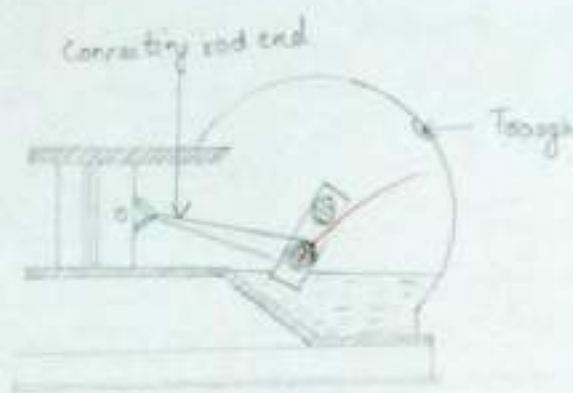
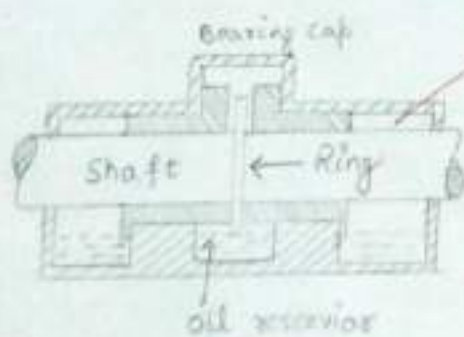


FIGURE-1



Splash Lubrication System.



Ring Oiling System



M. B. M. UNIVERSITY JODHPUR

Date 6/4/23

Experiment No. 3

P. No. 21

Name Zahid Khan

Faculty No. 20/24765

Class B.E. II Sem Batch. A4

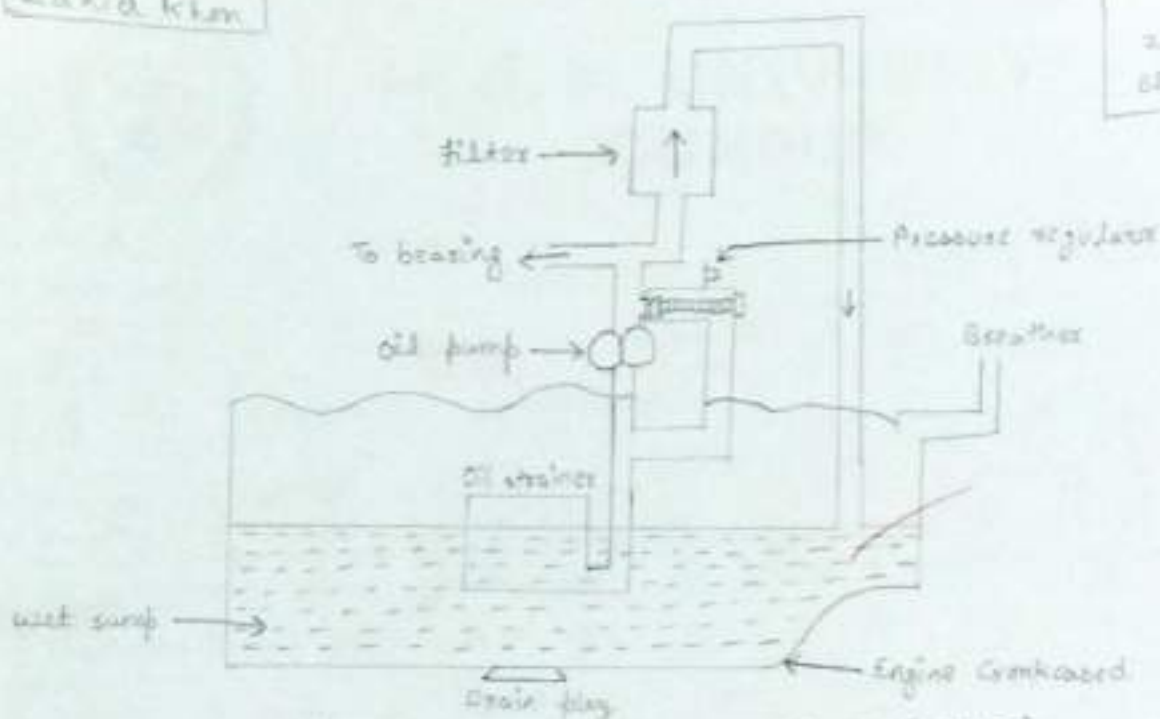
It thus directs the oil through holes in the caps to the big end bearings. Due to splash of oil it reaches the lower position of the cylinder walls, crankshaft & other parts required lubrication. Surplus oil eventually flows back to the oil sump. Oil level in the lubrication throughs is maintained by means of a oil pump which takes oil from sump, through a filter.

- Splash system is suitable for low & medium speed engines having moderate bearing load pressures. For high performance engines, which normally operate at high bearing pressure & rubbing speeds the system does not serve the purpose.

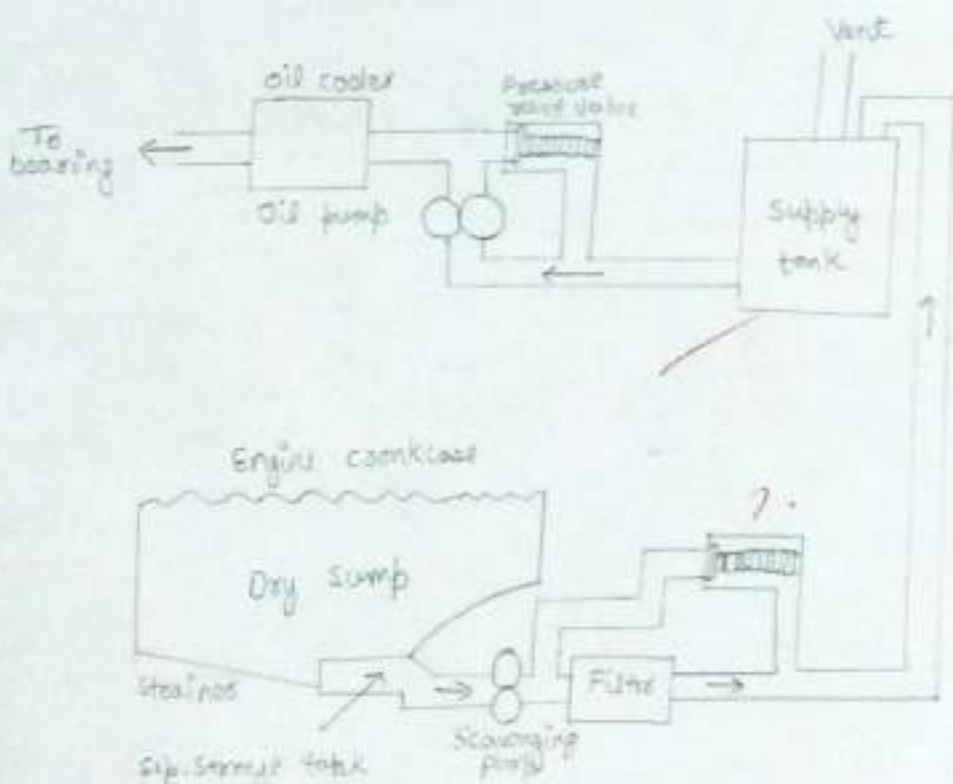
(2) Dry sump lubrication system ^(wing type lubricator) (For high Capacity engines):-

In this system, the oil from the sump is carried to a separate storage tank outside the engine cylinder block. The oil from sump is pumped by means of a pump through filters to the storage tank. Oil from storage tank is pumped to the engine cylinder through oil cooler. Oil pressure may vary from 3 to 8 bar.

- Dry sump lubrication system is generally adopted for high capacity engines.
- This type of lubrication is more useful for medium speed engines because at high speeds the oil will be thrown



Wet sump lubrication system (WSLS)



Dry Sump lubrication system (DSLS)



M. B. M. UNIVERSITY JODHPUR

Date: 3/4/23 Experiment No. 3 P. No. 23
Name: Zahid Khan Faculty No. 20/24765 Class: BE VI Sem Batch: 14

off due to centrifugal force & at low speeds, the amount of oil carried is inadequate.

(3) Forced / Pressured system :- (a) semi-pressure system

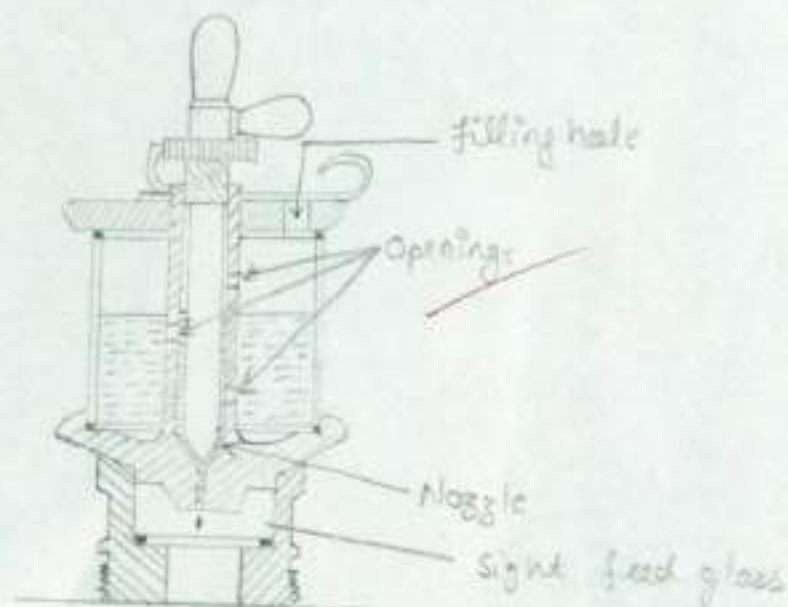
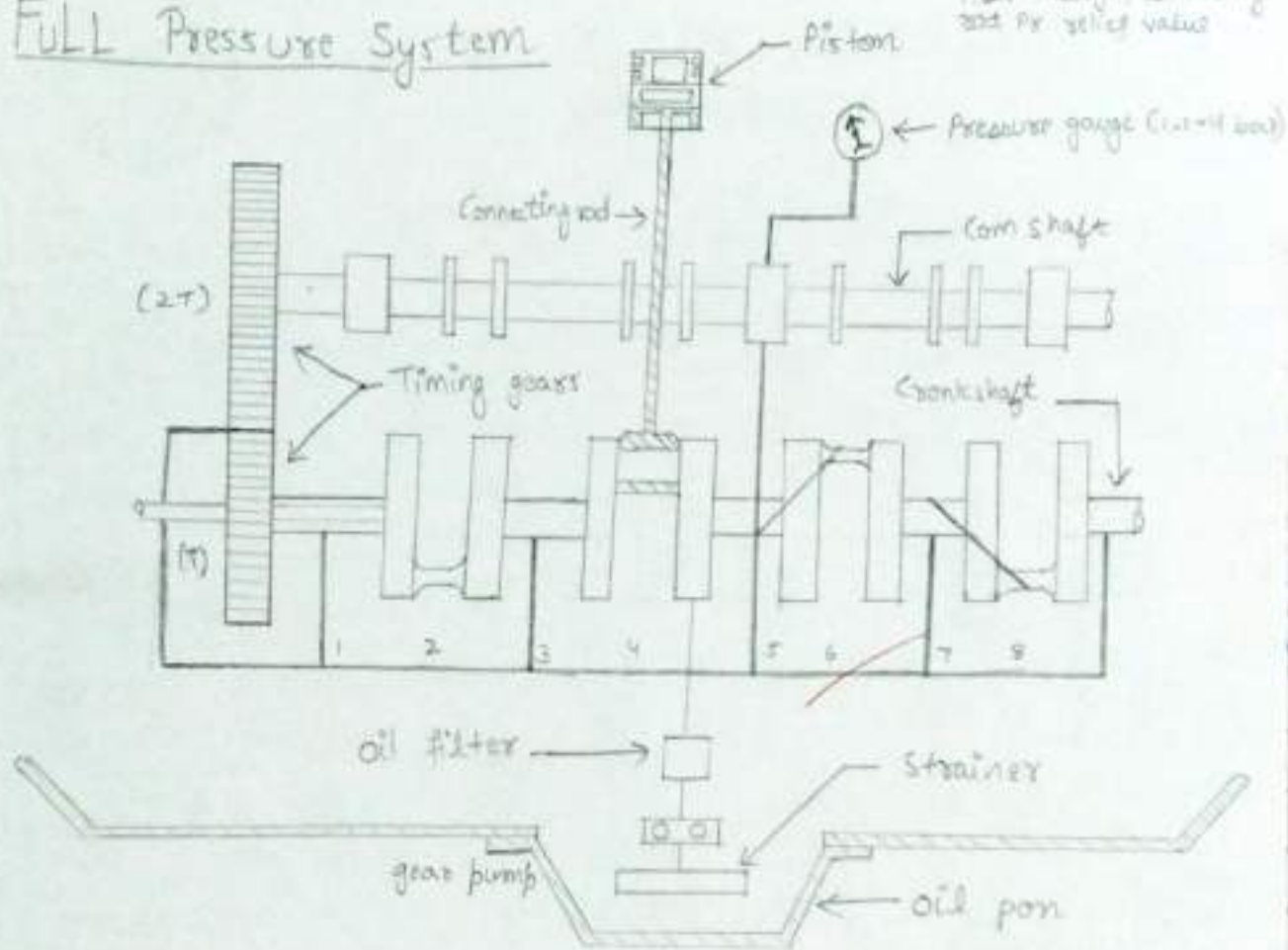
- This method is a combination of splash & pressure systems. It incorporates the advantages of both. In this case main supply of oil is located in the base of crank chamber. Oil is drawn from the lower portion of the sump through a filter & is delivered by means of a gear pump at pressure of about 1 bar to the main bearings. The big end bearing are lubricated by means of a spray through nozzles. Thus oil also lubricates the cams, crankshaft bearings, cylinder walls & timing gears. An oil pressure gauge is provided to indicate satisfactory oil supply.
- The system is less costly to install as compared to pressure systems. It enables higher bearing loads & engine speeds to be employed as compared to splash system.

(b) Full pressure system :-

- In this system, oil from oil sump is pumped under pressure to the various parts requiring lubrications. The oil is drawn from the sump through filter & pumped by means of a gear pump. Oil is delivered by the pressure pump at pressure

FULL Pressure System

Hole through connecting rod for relief valve



Drop feed oil cup



M. B. M. UNIVERSITY JODHPUR

Date 6/4/23 Experiment No. 3 P. No. 25
Name Zahid Khan Faculty No. 20/24765 Class BE II Sem Batch M

working from 1.5 bar to 4 bar. The oil under pressure is supplied to main bearings of crankshaft & crankshaft. Holes drilled through the main crankshaft bearing journals, communicate oil to the big end bearings & also small end bearings through hole drilled in connecting rods. A pressure gauge is provided to confirm the circulation of oil to the various parts. A pressure regulating valve is also provided on the delivery side of this pump to prevent excessive pressure.

- This system finds favour from most of the engine manufacturers as it allows high bearing pressure & rubbing speeds.
- The general arrangement of wet sump lubrication system is shown in figure. In this case oil is always contained in the sump which is drawn by the pump through a strainer.

(4) Lubrication of cylinder & small end bearing of connecting rod/Drop feed lubricator :-

- The cylinder, small end bearing (gudgeon pin), valve gear pins, rocker shaft, crank pins etc. are lubricated by drip system.
- In drip system oil is fed to the machine parts drop by drop, from an oil cup. Although it is not an efficient



M. B. M. UNIVERSITY JODHPUR

Date: 6/4/23 Experiment No. 3 P. No. 26
Name: Zahid Khan Faculty No. 20/24765 Class: DE VII Sem Batch: A4

method yet it is often the most convenient way of lubricating the external parts of engines & machines. Valve gear pins, rocker shafts, main bearings of small engines, crank pins, crosshead pins, line shaft bearings, & many other machine & engine parts are lubricated in this fashion.

- It comprises of an oil reservoir or cup made of glass. At the top of the cup there is an opening called filling hole through which the lubrication oil is poured into the cup. The oil enters into the inner chamber through the openings & it falls drop by drop due to gravity by the nozzle. The passage of oil through the nozzle is controlled by a needle valve. A sight feed glass is provided to see the drop of oil.

(v) Mixing lubrication:- (mist lubrication)

- Mixing lubricant with fuel is a common method of providing lubrication to 2-stroke engines. In 2-stroke engines, lubrication is typically achieved by adding oil to the fuel mixture, which is then burned along with the fuel during the combustion process.
- The oil helps lubricate the moving parts of the engine, such as crankshaft, piston & connecting rod. It also helps



M. B. M. UNIVERSITY JODHPUR

Date 6/4/23 Experiment No. 3 P. No. 27
Name Zahid Khan Faculty No. 20/24765 Class BE II Sem Batch 14

cooling the engine by carrying away heat from the combustion process.

- It is important to use the correct oil-to-fuel ratio as specified by the engine manufacturer, as using too little oil can result in insufficient lubrication & engine damage, while using too much oil can result in fouled spark plugs & poor engine performance.
- One more example of this is steam engines.

11/10/23



M. B. M. UNIVERSITY JODHPUR

Date 6/4/23

Experiment No. 4

P. No. 28

Entha

Name Zahid Khan

Faculty No. 20/24765

Class B.E. III Sem. Batch AY

* Objective:- To study engine starting motors (DC motor) & driving arrangement (Bendix drive).

* Apparatus required:- DC motor & Bendix drive mechanism

* Theory:-

(a) Engine starting motors:-

- The starting motor, is an important component of an internal combustion engine. It is responsible for starting the engine by turning the crankshaft until the engine reaches its self-sustaining speed. The most common type of starting motor used in automobile applications is a DC motor.

- Here are some key features & components of a DC starting motor:-

1. Armature:- It is the rotating part of DC motor. It consists of a core made up of iron laminations & a series of winding that are connected to commutator.
2. Commutator:- It helps to reverse the dirxn of current in the armature windings as the motor rotates. This is necessary to maintain the dirxn of rotation of the motor.
3. Brushes:- The brushes are 2 small piece of conductive material that are used to transfer current from the stationary part



M. B. M. UNIVERSITY JODHPUR

Date: 6/4/23 Experiment No. 4 P. No. 29
Name: Zahid Khan Faculty No. 20/24765 Class: AE IV Sem Batch: Ay

- of the motor (to stator) to the rotating part (the armature).
4. Field coils:- They are located on the stator & are used to create a magnetic field that interact with the magnetic field created by the armature. This interaction creates a torque necessary to turn the engine.
5. Solenoid:- It is an electromechanical device that engages the starter motor to the engine flywheel when the driver turns the ignition key.

- When the driver turns the ignition key, current flows from the battery to the solenoid, which engage the starter motor to the engine flywheel. The armature start to rotate, and the commutator & brushes helps to maintain the direction of rotation. The magnetic field created by the field coils interacts with the magnetic field created by the armature, creating the necessary torque to turn the engine.
- Once the engine starts running, the starter motor disengages from the flywheel, & the engine takes over the job of keeping itself running.

• some important points:-

1. For series motor, if mesh $\eta = 85\%$ then $BP = 8.5 \text{ kW}$ & $FP = 1.5 \text{ kW}$.
2. If a 12 V Battery is used then $Amp = \frac{15 \times 1000}{12} = \frac{150}{12} = 12.5 \text{ Amp}$.